A fixed-target approach to the sharp iid Berry–Esseen inequality*

Bartosz Kołodziejek Kacper Rodziewicz

30 October 2026

Abstract

The conjectured sharp iid Berry–Esseen constant is $C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$. Under the structural input and finite-verification conditions stated in this paper, we show that the sharp inequality for every sample size through 699 implies the sharp inequality for all sample sizes, strictly above 699. The finite base is not proved here.

The argument studies a law and threshold maximizing the excess above the target C_{E} at a fixed sample size. The first-order optimality condition and zero-bias integration give an exact identity involving a nonnegative support-gap defect. Its lattice specialization is a moment–span inequality, and its zero case selects Esseen's extremal two-point law. Exact rational localization, using the sharp bound at the preceding index, bounds this defect and the threshold. For $n \geq 700$, the resulting law lies in an explicit region around that two-point law. A direct comparison with a binomial sum then gives strictly negative excess for every law in this region. This local comparison is a separate result with explicit support and noise bounds; it does not require extremality or the predecessor inequality.

Contents

1 The problem, results, and proof strategy 2
2 Fixed-target extremizers and contact 5
3 The two-point skeleton and its moment defect 8
4 Conservation and control of the defect 12
5 Quantitative localization 15
6 From small defect to a common support region 16
7 The direct comparison with a binomial law 21
8 Composition and the finite boundary 30
9 Discussion 31
A A uniform Fourier enclosure with explicit input hypotheses 32
B Elementary Gaussian constants 37
C The identity without contact 39
D Details of the coarse localization 39
E Certificates and independent reproduction 52

*This paper was written with substantial assistance from AI systems; see the end of Section 9.

1 The problem, results, and proof strategy

1.1 The problem and the candidate constant

Let $\mathcal{P}$ be the laws on $\mathbb{R}$ with mean zero, variance one and finite third absolute moment $\rho(P) = \mathbb{E}|X|^3$. For independent copies $X_1, \dots, X_n$ of P, put $S_n = \sum_i X_i$ and

$$\Delta_n(P) = \sup_{H \in \mathbb{R}} |\mathbb{P}_P(S_n \le H) - \Phi(H/\sqrt{n})|, \qquad C_n = \sup_{P \in \mathcal{P}} \frac{\sqrt{n}\,\Delta_n(P)}{\rho(P)}.$$

Here Φ and ϕ are the standard normal distribution function and density. The sharp-constant conjecture is

$$\sup_{n \ge 1} C_n = C_{\mathrm{E}}, \qquad C_{\mathrm{E}} = \frac{3 + \sqrt{10}}{6\sqrt{2\pi}} = 0.409732\ldots. \tag{1}$$

Throughout, H is a threshold for the unnormalized sum S_n, and a terminating decimal in an inequality denotes the corresponding exact rational number. Affine standardization gives the equivalent result for any mean and positive variance, with third absolute central moment divided by the cube of the standard deviation.

Berry and Esseen established the classical order $n^{-1/2}$ [3–5]. Esseen's lower bound [6] explains the proposed constant. If $p_* = (4 - \sqrt{10})/2$, $q_* = 1 - p_*$ and $X = (B - p_*)/\sqrt{p_* q_*}$ for $B \sim \mathrm{Bernoulli}(p_*)$, the fixed-law lattice limit gives

$$\lim_{n \to \infty} \frac{\sqrt{n}\Delta_n(P)}{\rho(P)} = \frac{3 + q_* - p_*}{6\sqrt{2\pi}(p_*^2 + q_*^2)} = C_{\mathrm{E}}. \tag{2}$$

Indeed $p_*^2 + q_*^2 = 10 - 3\sqrt{10}$ and $(3 + q_* - p_*)/(p_*^2 + q_*^2) = 3 + \sqrt{10}$. Every constant below C_{E} therefore fails for a sufficiently large sum of this one law. The upper conjecture is associated with Zolotarev [19, 20]; Schulz proves the strict finite-index bound for every nondegenerate Bernoulli law [11].

1.2 Results and their hypotheses

The results have three distinct roles. We state their interfaces here; the cited theorems give their complete hypotheses and proofs.

Result	Conclusion and required inputs
Local comparison, Theorem 7.1	For $n \ge 700$, every law in its specified conditional-support region, with $p \in [3/8, 93/200]$, $0 < \lambda \le 7/20$ and $\lvert H\rvert \le 46/25$, has excess below $-\lambda/100$. The zero-noise excess is strictly negative. This statement uses only its explicit law hypotheses.
Localization, Theorem 5.1	A maximizer with nonnegative excess at $n \ge 42$ satisfies $nD \le 32.20665378$ and $\lvert H\rvert \le 1.83605680$, under $C_{n-1} \le C_{\mathrm{E}}$, the structural inequality and the two stated finite-verification conditions.
Global implication, Theorem 8.2	With the retained structural, binomial and finite-verification inputs, a base $C_j \le C_{\mathrm{E}}$ for every $j \le 699$ implies $C_n < C_{\mathrm{E}}$ for $n \ge 700$ and $\sup_n C_n = C_{\mathrm{E}}$. The finite base remains open here.

The structural input is displayed in (50); the finite-binomial input is (29). Conditions D.3 and D.5 specify the finite localization checks. Their exact replay is recorded in Appendix E. The analytical reduction, the whole-domain coverage, and the correspondence of the checking code with those inequalities are the components of this computational proof. We retain these premises explicitly in the theorem statements.

1.3 Why the defect controls a maximizer

At a fixed n, maximize the difference between the two sides of the candidate inequality:

$$a(P,H) = \sqrt{n}\{\mathbb{P}_P(S_n \le H) - \Phi(H/\sqrt{n})\} - C_{\mathrm{E}}\rho(P). \tag{3}$$

Proposition 2.1 proves attainment. Its first-order condition bounds $y \mapsto \mathbb{P}(S_{n-1} \le H - y)$ by an explicit piecewise cubic function that agrees with it on the closed support. This contact condition allows us to use the geometry of that support.

Let $\tau(t) = \mathbb{E}[X\mathbf{1}_{\{X \ge t\}}]$ be the zero-bias density and $J = \sum_g \tau_g |g|^2$, where g runs over bounded support gaps and τ_g is the constant value of τ there. A lattice law of span h has $J \ge h$; if every successive gap has length h, then $J = h$. Thus J measures gap size with zero-bias weights. The inequality

$$\gamma + 3J \le (3 + \sqrt{10})\rho, \qquad \gamma = \mathbb{E}X^3,$$

proved in Lemma 3.1, extends the moment–span bound from lattice laws to arbitrary standardized laws. Its slack

$$D = (3 + \sqrt{10})\rho - \gamma - 3J$$

is nonnegative and vanishes exactly at the positive-skew Bernoulli law in (2). The lattice consequence and its reflection are proved in Corollary 3.3.

Integrating the contact relation against τ in two ways gives the conservation identity of Theorem 4.1. Since $C_{\mathrm{E}} = (3 + \sqrt{10})\phi(0)/6$, its two leading moment terms combine into a multiple of D. Solving for the excess gives

$$a = \frac{\mathcal{R}_c - nC_{\mathrm{E}}\rho(1-e) - n\phi(0)eD/6 - n^{3/2}L}{n + H^2}, \qquad e = \exp(-H^2/(2n)).$$

Here $L \ge 0$ is a weighted contribution from the law of $H - S_{n-1}$ inside the support gaps, and $\mathcal{R}_c$ is an explicitly defined remainder. The three subtracted terms are nonnegative. A nonnegative excess therefore requires $n\phi(0)eD/6 \le \mathcal{R}_c$. Once localization bounds both $\mathcal{R}_c$ and $|H|$, this bounds nD. The uniform bounds come from Theorem 5.1, not from a fixed-law asymptotic expansion.

The predecessor hypothesis supplies an upper bound for $\mathbb{P}(S_{n-1} \le H - t)$ in the contact condition, which excludes remote support points; see (52). This support argument is used both in coarse localization and in the final two-ray estimate. With $nD \le 32.20665378$, the geometry at $n \ge 700$ gives $3/8 < p < 93/200$, inward support displacement below $1/250$ and $\lambda < 7/20$. Count conditioning then compares the law with a binomial sum. One threshold count controls noise up to $1/100$; five neighboring counts control the remaining interval. Both comparisons keep all probabilities entering from other counts.

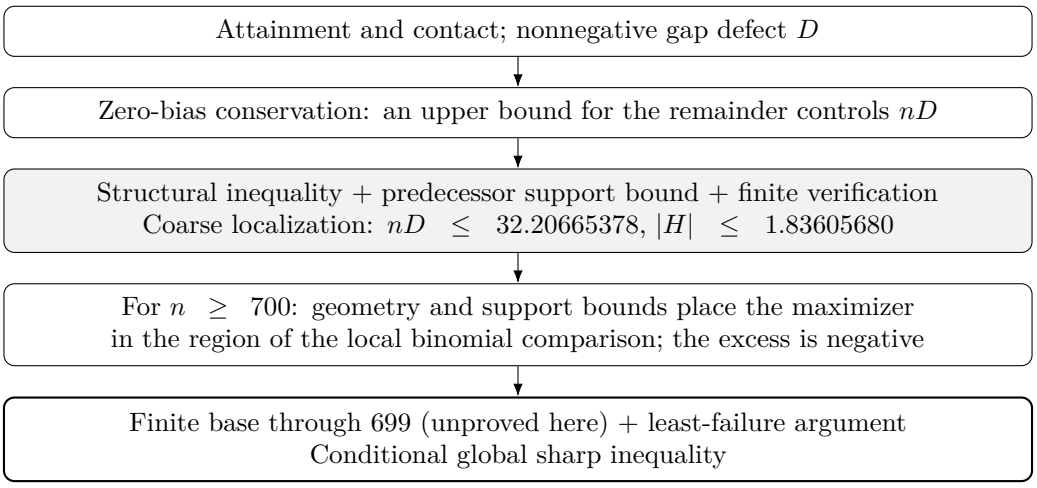


Figure 1: The proof's dependencies. The shaded step uses the specified finite verification; the final step also requires the unproved finite base. The local comparison is a theorem for its explicit law region, independently of the preceding localization.

1.4 The computer-assisted localization

Analysis restricts a possible maximizer to a bounded moment region. It also bounds the remainder by functions of moment and support parameters. The computation encloses these functions on finitely many closed boxes. Each box is either excluded by a necessary inequality, or assigned bounds valid throughout that box. The boxes cover their parent regions, including shared boundaries, so every possible maximizer retains a valid bound.

The certificate is a compact record of this process. It stores checkpoints for 24 third-moment cells over four signed-moment passes, including support radii, bounds and descendant counts; the individual 11264 nominal boxes are reconstructed. Two preliminary coarse passes are reconstructed as well. The exporter chooses a successful sequence of passes and rounds its reported bounds outward. The acceptor recomputes the prescribed subdivisions and verifies their inequalities and output bounds.

All numerical acceptance uses integers and exact rational numbers, with directed rational enclosures for roots, exponentials and Gaussian constants. The exploratory program and the checker share the routine `signed_core.stage`; acceptance is an exact replay of that calculation. Independent checking requires auditing these routines against the mathematical rules or implementing those rules separately. Appendix E gives a code-reading guide, reproduction commands and the recorded execution environment.

Subdivision and rigorous enclosure have established precedents in computer-assisted proofs [15]. Numerical optimization also appears in Berry–Esseen estimates [12, Section 2]. The custom JSON format organizes this calculation; the rigor comes from the valid enclosures, coverage and correct implementation of the rules.

1.5 Related work

Fourier and structural moment inequalities progressively improved the universal upper bound [9, 12, 14, 16, 17]. We use Shevtsova's structural inequality explicitly in localization. Recent manuscripts report smaller upper constants [8, 18], and He and Cheng report eventual sharpness beyond a uniform, very large threshold [7]. Work on independent arrays and extremizers addresses the related non-iid problem [1, 2]. These recent reported results are not used in our proof. An earlier AI-assisted working draft is on Hexagon [10].

1.6 Organization and notation

Section 2 gives attainment, contact and reflection. The next two sections develop the defect and the conservation identity; Sections 5 and 6 bound and localize a possible maximizer. Section 7 proves the independent local comparison, and Section 8 combines these results. The appendices contain the Fourier bounds, rational constants, off-contact identity, full localization calculation, and reproduction instructions.

The following table distinguishes the three scales used in the proof. Gap-specific parameters are introduced after the existence of a central gap has been established.

Symbol	Meaning and scale
H, $H/\sqrt{n}$	Threshold for the original sum; its normal-standardized value.
$B_0 = \mathbf{1}_{\{X>0\}}$, p, q	Sign class and its probabilities for the original marginal.
m, $\ell = m/(pq)$	$m = \mathbb{E}X_+ = \mathbb{E}X_-$; distance between conditional means.
W, $\theta = \mathbb{E}W^2$	$W = X - \mathbb{E}(X \mid B_0)$; conditional noise in original units.
$U = W/\ell$, $s = \mathbb{E}U^2$, $\lambda = ns$	Noise in count units; marginal and total noise variances.
$t = np + H\sqrt{pq+s} = k + u$	Count threshold, integer anchor, and phase.
a_0, a_1, $d = a_0 + a_1$	Inward support displacements in count units.
J, D	Weighted support-gap sum and defect of the original law.

The exact normalization is proved in (6). It identifies the count-coordinate law with the original X and leaves H as its raw sum threshold. The signed moments are $\gamma = \mathbb{E}X^3$ and $\beta = \mathbb{E}[X|X|]$.

2 Fixed-target extremizers and contact

We maximize the difference between the two sides of the desired inequality. Its sign detects failure of the target bound, while the coefficient of the third absolute moment is fixed in advance. Throughout, sums have independent identically distributed summands with mean zero, variance one and finite third absolute moment ρ. Fix $n \geq 2$, $c > 0$, and a real raw sum threshold H. Set

$$z = H/\sqrt{n}, \quad F = \mathbb{P}(S_n \leq H), \quad a = \sqrt{n}\{F - \Phi(z)\} - c\rho,$$
$$\phi_z = \phi(z), \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3.$$

In particular a is not divided by ρ.

2.1 Attainment, contamination and the entire closed support

Proposition 2.1. *For each fixed n and $c > 0$, the supremum of a over these laws and thresholds is attained, and the closed-CDF discrepancy at every attaining pair is positive. At every maximizing pair,*

$$I(y) = n^{3/2}\{h(y) - F\} + ny\phi_z + \tfrac{1}{2}H\phi_z(y^2-1) + \tfrac{3}{2}c\rho(y^2-1) - c\{|y|^3 - \rho - 3y\beta\} \leq 0, \qquad h(y) = \mathbb{P}(S_{n-1} \leq H - y), \tag{4}$$

with equality on the whole closed support. That support is compact. Neither statement assumes $a \geq 0$ or $H \geq 0$.

Equivalently, $\Delta = Q - h \geq 0$ everywhere and $\Delta = 0$ on the closed support, where

$$Q(y) = F + \frac{H\phi_z + c\rho}{2n^{3/2}} - \frac{n\phi_z + 3c\beta}{n^{3/2}}y - \frac{H\phi_z + 3c\rho}{2n^{3/2}}y^2 + \frac{c}{n^{3/2}}|y|^3. \tag{5}$$

The contact condition is a first-order optimality condition: replacing one summand by a fixed value y gives the conditional probability $h(y)$. After recentering and rescaling the perturbed law, the directional change of the excess is nonpositive. Its average against P is zero, which forces equality on the closed support. The function Q is piecewise cubic because it contains $|y|^3$.

Proof. Write a^* for the supremum. Since $\rho \geq 1$ and the discrepancy is at most one, $a^* \leq \sqrt{n}-c$. Symmetric Bernoulli evaluated at $H = n$ gives $a^* \geq \sqrt{n}(1 - \Phi(\sqrt{n})) - c > -c$. For a maximizing sequence, $a_j \to a^*$ and $c\rho_j \leq \sqrt{n} - a_j$ give a common third-moment ceiling. The discrepancies satisfy $\sqrt{n}(F_j - \Phi(H_j/\sqrt{n})) = a_j + c\rho_j \geq a_j + c > \eta$ eventually for some $\eta > 0$. Chebyshev bounds the negative threshold tail by n/H_j^2; the positive discrepancy is at most the normal tail, also at most n/H_j^2. Thus H_j is bounded.

Tightness gives a subsequence $P_j \Rightarrow P$, $H_j \to H$ and $\rho_j \to b$. The common third-moment ceiling M gives tail bounds $\int_{|x|>K} x^2 dP_j \leq M/K$ and $\int_{|x|>K} |x| dP_j \leq M/K^2$. Bounded continuous truncations therefore preserve mean and variance. Lower semicontinuity gives $1 \leq \rho(P) \leq b$. For fixed n, the sum laws converge weakly. Applying the closed-set inequality first at $H + t$ and then letting $t \downarrow 0$ gives $\limsup F_j(H_j) \leq F_P(H)$, including an atom at H. Consequently

$$a(P, H) \geq a^* + c\{b - \rho(P)\} \geq a^*.$$

Maximality forces equality and $\rho(P) = b$. Thus third moments converge along this maximizing sequence. The discrepancy remains positive since $a^* + c\rho \geq a^* + c > 0$.

The contact inequality is a first-variation condition. To obtain it, contaminate the raw marginal by $(1 - \epsilon)P + \epsilon\delta_y$. Its mean is ϵy, variance $v_\epsilon = 1 + \epsilon(y^2 - 1) - \epsilon^2 y^2$, and centered third absolute moment has derivative $|y|^3 - \rho - 3y\beta$ at zero. The last assertion follows by dominated differentiation of $|x - \epsilon y|^3$, with bound $3|y|(|x| + |y|)^2$. Evaluate the standardized law at the standardized image of the fixed raw event $S_n \leq H$. The raw event probability is a polynomial in ϵ, with derivative $n(h(y) - F)$ even at an atom collision. The normal argument is $(H - n\epsilon y)/\sqrt{nv_\epsilon}$, whose derivative is $-\sqrt{n}y - H(y^2 - 1)/(2\sqrt{n})$. The one-sided derivative of the excess is exactly (4); maximality makes it nonpositive.

Its integral against P is zero. All terms are integrable. The closed-event map h is upper semicontinuous; at an atom it takes the larger one-sided value. Thus I is upper semicontinuous. Strict negativity at a closed-support point would hold with a uniform negative margin on a neighborhood of positive P-mass, contradicting its zero integral. Finally $I(y) \leq -c|y|^3 + C_2 y^2 + C_1|y| + C_0$ for constants depending on this pair. It is strictly negative on both sufficiently distant rays, proving compact support. □

2.2 Reflection, equality, and the first unresolved index

The sign of the attained maximum detects failure of the strict bound $C_n < c$, including equality. If C_n denotes the supremum of the normalized absolute discrepancy, reflection gives, including an atom at H,

$$\mathbb{P}(-S_n \leq -H) - \Phi(-z) = \Phi(z) - \mathbb{P}(S_n < H) \geq \Phi(z) - F.$$

Thus each negative closed-CDF error is bounded by a positive closed-CDF error of the reflected law; equality at an atom is not assumed. Consequently C_n is also the supremum of positive closed scores r. Since $a = \rho(r - c)$, $C_n < c$ gives $a^* \leq C_n - c < 0$, and $C_n > c$ gives $a^* > 0$. If $C_n = c$, choose $r_j \uparrow c$ with $r_j \geq c/2$. Then $\rho_j \leq 2\sqrt{n}/c$, so $\rho_j(r_j - c) \to 0$ and $a^* = 0$. Hence $C_n \geq c$ supplies an attained pair with $a \geq 0$.

Corollary 2.2 (First failure of a strict bound)**.** *Assume a genuine finite base $C_j \leq c$ for $1 \leq j < N$. If a strict bound $C_n < c$ fails at any index $n \geq N$, there is a least such index, an*

attained fixed-target pair with $a \geq 0$, and predecessor bounds $C_j \leq c$ for every $j < n$. The bounds are strict for $N \leq j < n$.

Proof. The nonempty set of failing indices at least N has a least member. The preceding equivalence supplies its maximizing pair. Indices below N use the finite base; the other earlier indices use minimality. □

Thus an argument excluding every such pair for $n \geq N$ would give strict bounds above the finite base, including exclusion of equality. The subsequent entry argument uses only $C_{n-1} \leq c$; it does not use an upper bound for the actual score $c + a/\rho$. The finite base and the eventual exclusion remain separate obligations.

2.3 Two elementary moment domains

These two inequalities restrict the signed moments in terms of ρ. They hold for every standardized law with finite third moment, including laws of unbounded support.

Theorem 2.3 (Signed and absolute third moments)**.** *If $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, $\rho = \mathbb{E}|X|^3 < \infty$, and $\gamma = \mathbb{E}X^3$, then*

$$\rho \geq \frac{\gamma^2 + 2}{\sqrt{\gamma^2 + 4}}.$$

This is the sharp moment relation of Shevtsova [13]; in the cited arXiv version its equivalent form is Theorem 2. The following factorization gives a self-contained proof.

Proof. Set $s = \sqrt{\gamma^2 + 4}$, $a = (s - \gamma)/2$, $b = (s + \gamma)/2$ and $c_\gamma = \gamma(\gamma^2 + 6)/s^3$. Then $a, b > 0$, $ab = 1$ and $a + b = s$. Define

$$w(y) = |y|^3 - c_\gamma y^3 - \frac{12y^2}{s^3} + \frac{6\gamma y}{s^3} + \frac{4}{s^3}.$$

For $y, t \geq 0$, direct multiplication gives

$$w(y) = \frac{2a^2}{s^3}(y - b)^2\{2 + (a + 3b)y\}, \qquad w(-t) = \frac{2b^2}{s^3}(t - a)^2\{2 + (3a + b)t\}.$$

Thus $w \geq 0$ on the whole real line. Taking expectations and using the first three signed moments gives

$$0 \leq \mathbb{E}w(X) = \rho - c_\gamma\gamma - 8/s^3 = \rho - (\gamma^2 + 2)/s.$$

All terms are integrable under the stated third-moment hypothesis. □

Lemma 2.4 (The signed second moment)**.** *With the same normalization and $\beta = \mathbb{E}[X|X|]$,*

$$|\beta| < 1, \qquad \rho \geq \frac{1 + \beta^2}{\sqrt{1 - \beta^2}}.$$

Proof. Both signs have positive probability. Put $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$ and $a_\pm = \mathbb{E}X_\pm^2 = (1 \pm \beta)/2 > 0$. If $p_\pm = \mathbb{P}(\pm X > 0)$, Cauchy's inequality gives $m^2 \leq p_\pm a_\pm$. Since $p_+ + p_- \leq 1$, this implies

$$m^2(1/a_+ + 1/a_-) \leq 1, \qquad 2m \leq \sqrt{1 - \beta^2}.$$

Another application of Cauchy's inequality, to $X_\pm^{1/2}$ and $X_\pm^{3/2}$, gives $m\mathbb{E}X_\pm^3 \geq a_\pm^2$. Summing yields $m\rho \geq (1 + \beta^2)/2$, which proves the claim. A possible zero atom enters only through $p_+ + p_- \leq 1$ and causes no exception. □

3 The two-point skeleton and its moment defect

Every centered law can be separated into its positive and nonpositive parts. Replacing each part by its conditional mean gives a two-point law; the remaining conditional fluctuations measure its departure from that law. We first make this decomposition exact, then identify a moment defect that vanishes at Esseen's extremal distribution.

3.1 The sign split and its units

Fix $P \in \mathcal{P}$ and put

$$p = \mathbb{P}(X > 0), \quad q = 1 - p, \quad v = pq, \quad m = \mathbb{E}X_+ = \mathbb{E}X_- > 0, \quad B_0 = \mathbf{1}_{\{X>0\}}.$$

Both p, q are positive; a possible atom at zero is assigned to the nonpositive class. Define $V = \mathbb{E}[X \mid B_0]$ and $W = X - V$, so that

$$V = -m/q \quad (B_0 = 0), \qquad V = m/p \quad (B_0 = 1), \qquad \theta = \mathbb{E}W^2 = 1 - m^2/v < 1.$$

Write $\theta_- = \mathbb{E}[W^2; B_0 = 0]$ and $\theta_+ = \mathbb{E}[W^2; B_0 = 1]$. The distance between the two conditional means is

$$\ell = m/v = \sqrt{(1-\theta)/v} > 0.$$

Set $U = W/\ell$, $Y = B_0 + U$, and $s = \mathbb{E}U^2$. Then

$$X = \ell(Y - p) = \frac{B_0 + U - p}{\sqrt{pq + s}}, \qquad \mathbb{E}[U \mid B_0] = 0, \tag{6a}$$

$$1 = pq\ell^2 + \theta, \qquad s = \theta/\ell^2, \qquad v + s = \ell^{-2}. \tag{6b}$$

Let U_0, U_1 have the two conditional laws, and put

$$w_i = \mathbb{E}U_i^2, \quad \mu_i = \mathbb{E}U_i^3, \quad s = qw_0 + pw_1 = \frac{v\theta}{1-\theta}, \quad \lambda = ns.$$

The moments μ_i are finite because $\rho < \infty$. Higher moments will be used only after bounded support is known: for extremizers this follows from Proposition 2.1, and for the local comparison it is an explicit hypothesis.

If $s = 0$, both nonnegative conditional variances vanish, so $U_0 = U_1 = 0$ almost surely. This is exactly the standardized Bernoulli law. If $s > 0$, the parameter $\alpha = pw_1/s$ lies in $[0, 1]$ and gives $w_0 = s(1-\alpha)/q$, $w_1 = s\alpha/p$.

The signs of X are still those defining the two classes, so expansion of the actual conditional moments gives

$$\begin{aligned}
\rho/\ell^3 &= v(1-2v) + 3v(w_0 + w_1) - q\mu_0 + p\mu_1, \\
\gamma/\ell^3 &= v(q-p) + 3v(w_1 - w_0) + q\mu_0 + p\mu_1, \\
\beta/\ell^2 &= v(q-p) + pw_1 - qw_0.
\end{aligned} \tag{7}$$

For example, on the nonpositive class $|X|^3 = \ell^3(p - U_0)^3$; on the positive class it is $\ell^3(q + U_1)^3$. Conditional centering removes the linear terms and gives the first identity; the other two follow in the same way. The three identities share the same pair (μ_0, μ_1).

3.2 A nonnegative moment defect

The constant $3+\sqrt{10}$ has a simple geometric explanation. The signed third moment and the first absolute moment satisfy a circle inequality; the lengths of the support gaps are controlled by the same first absolute moment. Neither fact requires bounded support.

Define

$$\tau(t) = \mathbb{E}[X\mathbf{1}_{\{X\geq t\}}], \qquad \sigma = \sqrt{10}.$$

For $t<0$, centering gives $\tau(t) = -\mathbb{E}[X\mathbf{1}_{\{X<t\}}] \geq 0$; for $t\geq 0$ nonnegativity is immediate. Signwise Tonelli integration gives

$$\int_{\mathbb{R}} \tau(t)\,dt = 1, \qquad 2\int_{\mathbb{R}} |t|\tau(t)\,dt = \rho. \tag{8}$$

On each bounded complementary interval (a,b) of the closed support, τ is constant in the interior, with value $\tau_{a,b}$. Put

$$J = \sum_{(a,b)} \tau_{a,b}(b-a)^2, \qquad D = (3+\sigma)\rho - \gamma - 3J. \tag{9}$$

There are at most countably many gaps; the next proof shows that the nonnegative sum is finite. Endpoint atoms do not affect its integrals.

Lemma 3.1 (A moment circle and the gap inequality)**.** *For every $P\in\mathcal{P}$,*

$$\gamma^2 + (2m)^2 \leq \rho^2, \qquad J\leq \rho + 2m, \qquad \gamma + tJ \leq \bigl(t+\sqrt{1+t^2}\bigr)\rho \quad (t>0). \tag{10}$$

The last constant is sharp. Equality holds exactly at the standardized Bernoulli law with positive mass $p_t = (1+t-\sqrt{1+t^2})/2$. In particular, $D\geq 0$. If zero belongs to the support, then $D\geq(\sigma-1)\rho$; hence $D<(\sigma-1)\rho$ implies a central support gap $(-A_-,A_+)$ with $A_-,A_+>0$.

Proof. Write $a_\pm = \mathbb{E}X_\pm^2$ and $b_\pm = \mathbb{E}X_\pm^3$, so $a_+ + a_- = 1$, $b_+ + b_- = \rho$ and $b_+ - b_- = \gamma$. Cauchy–Schwarz on each sign class gives $a_\pm^2 \leq m b_\pm$. It also gives $m^2 \leq p a_+$ and $m^2 \leq q a_-$, where an atom at zero is included in q. Thus $m^2(1/a_+ + 1/a_-) \leq 1$, or $m^2 \leq a_+a_-$. Consequently $b_+b_- \geq (a_+a_-)^2/m^2 \geq m^2$, proving the circle inequality since $\rho^2-\gamma^2 = 4b_+b_-$.

Every gap on one side of zero satisfies $(b-a)^2 \leq 2\int_a^b |x|\,dx$. If a central gap $(-A_-,A_+)$ exists, its weight is m and its excess over this integral is $2mA_-A_+$. Moreover $0\leq \mathbb{E}[(X+A_-)(X-A_+)] = 1-A_-A_+$. Summing the gap bounds by Tonelli proves $J\leq\rho+2m$. If zero is in the support there is no central gap, so $J\leq\rho$. Finally Cauchy–Schwarz in $\mathbb{R}^2$ gives $\gamma + 2tm \leq \sqrt{1+t^2}\rho$, proving the family and the assertions about D.

Equality in the circle forces equality in the two signwise Cauchy bounds: the positive and negative parts are each concentrated at one nonzero value. Equality in $m^2\leq a_+a_-$ excludes mass at zero. For such a law, with positive mass p and $q=1-p$, one has $J=\rho+2m$. Equality in the last Cauchy bound requires $\gamma = 2m/t$, equivalently $t(1-2p) = 2p(1-p)$. Its unique solution in $(0,1)$ is the stated p_t. □

Corollary 3.2 (The zero-defect law)**.** *For $P\in\mathcal{P}$, $D=0$ if and only if*

$$X = (B_0 - p_*)/\sqrt{p_*q_*}, \qquad p_* = (4-\sqrt{10})/2, \quad q_* = 1-p_*, \quad B_0 \sim \text{Bernoulli}(p_*).$$

Proof. Apply the equality case of Lemma 3.1 with $t=3$. □

Corollary 3.3 (The lattice consequence)**.** *Let $P \in \mathcal{P}$ be a lattice law with maximal span $h > 0$. Then*

$$J \geq h, \qquad |\gamma| + 3h \leq (3 + \sqrt{10})\rho.$$

Equality in the second inequality holds exactly for the law of Corollary 3.2 and its reflection. If successive support points are all separated by h, then $J = h$.

Proof. The support is a closed subset of a translate of $h\mathbb{Z}$, so its points have Lebesgue measure zero and every bounded gap has length $k_g h$ for an integer $k_g \geq 1$. The density τ vanishes outside the convex hull of the support. Hence, by nonnegative integration,

$$\sum_g \tau_g k_g h = \int_{\mathbb{R}} \tau(t)\, dt = 1, \qquad J = \sum_g \tau_g k_g^2 h^2 \geq h.$$

These identities also hold for unbounded support: the gaps are countable and their contributions increase to the full integral. If all $k_g = 1$, the inequality is an equality. Under reflection the zero-bias density becomes $\tau(-t)$ almost everywhere. Thus reflection preserves ρ, J and h, and changes the sign of γ. Apply Lemma 3.1 to the orientation with nonnegative third moment. Equality forces $D = 0$, and the equality law has $J = h$, so Corollary 3.2 proves both directions. □

For a lattice law, J is therefore a weighted average of support-gap lengths, with weights $\tau_g|g|$ summing to one. The same functional is defined for arbitrary support, which is why it can replace the lattice span in the defect inequality.

In particular the defect separates two losses:

$$D = \underbrace{\sigma\rho - \gamma - 6m}_{\text{moment loss}} + 3\underbrace{(\rho + 2m - J)}_{\text{gap loss}},$$

each nonnegative. The point $(\gamma/\rho, 2m/\rho)$ lies in the upper unit disk. The line $x + 3y = \sqrt{10}$ supports that disk at $(1/\sqrt{10}, 3/\sqrt{10})$, explaining the coefficient in the moment loss. For a standardized Bernoulli law with positive mass p, direct calculation gives

$$J = \frac{1}{\sqrt{pq}}, \qquad \rho_B(p) = \frac{1 - 2pq}{\sqrt{pq}},$$
$$\gamma_B(p) = \frac{q - p}{\sqrt{pq}}, \qquad D_B(p) = \frac{(6 + 2\sigma)(p - p_*)^2}{\sqrt{pq}}. \tag{11}$$

Indeed its zero-bias density is $\sqrt{pq}$ on the interval between its two support points. The probability displacement therefore has a quadratic cost. The equality proof above is what rules out other laws with zero defect.

For quantitative estimates we need more information than their signs. Suppose a central gap $(-A_-, A_+)$ has been established, and write

$$T_+ = \mathbb{E}X_+^3 - mA_+^2, \quad T_- = \mathbb{E}X_-^3 - mA_-^2, \qquad R_+ = \mathbb{E}X_+^2 - mA_+, \quad R_- = \mathbb{E}X_-^2 - mA_-.$$

These four quantities are nonnegative. For the outer gaps put $J_o = J - m(A_- + A_+)^2$. A positive outer gap satisfies $(b - a)^2 \leq 2\int_a^b (t - A_+)\, dt$, so Tonelli gives

$$J_o^+ \leq 2\int_{A_+}^{\infty} (t - A_+)\tau(t)\, dt = \mathbb{E}[X_+(X_+ - A_+)^2] = T_+ - 2A_+R_+ \leq T_+.$$

Similarly $J_o^- \leq T_- - 2A_-R_- \leq T_-$. Expanding D, using $\sqrt{\sigma + 1}\sqrt{\sigma - 1} = 3$, gives

$$D = m\{\sqrt{\sigma + 1}A_- - \sqrt{\sigma - 1}A_+\}^2 + (\sigma - 1)T_+ + (\sigma + 1)T_-$$
$$+ 3(T_+ + T_- - J_o). \tag{12}$$

Every term is nonnegative. Cauchy–Schwarz also gives $1/(2\rho) \le m \le 1/2$: indeed $1 \le \rho\mathbb{E}|X| = 2m\rho$ and $\mathbb{E}|X| \le 1$. Since $\sigma - 1 > 2$, the decomposition yields $T_+ + T_- \le D/2$. If $D < 2$, then

$$m(A_-^2 + A_+^2) = \rho - T_+ - T_- \ge 1 - D/2, \qquad A_- + A_+ \ge \sqrt{2 - D}.$$

3.3 One joint budget for endpoint motion and noise

Let $\bar{A}_- = m/q = \ell p$, $\bar{A}_+ = m/p = \ell q$ be the conditional mean magnitudes. For a central support gap $(-A_-, A_+)$ define

$$\Delta_- = \bar{A}_- - A_-, \quad \Delta_+ = \bar{A}_+ - A_+, \qquad a_0 = \Delta_-/\ell, \quad a_1 = \Delta_+/\ell, \quad d = a_0 + a_1.$$

These quantities are nonnegative, $d < 1$, and $U_0 \le a_0$, $U_1 \ge -a_1$. The physical displacement $\Delta_- + \Delta_+$ is ℓd; the displacement d is in count units. Put $w = 1 - \theta = pq\ell^2$ and

$$\mathcal{Q}(p) = (\sigma + 1)p^2 + (\sigma - 1)q^2 - 6pq = (6 + 2\sigma)(p - p_*)^2.$$

Lemma 3.4 (Joint defect budget)**.** *For every $P \in \mathcal{P}$ whose closed support has a central gap $(-A_-, A_+)$ with $A_-, A_+ > 0$,*

$$\begin{aligned} D \ge{}& \ell w \mathcal{Q}(p) + 6\ell w\{d(1-d) + a_0 a_1\} \\ &+ \mathbb{E}[\{(\sigma - 1)(X + 2\bar{A}_+) + 6A_+\}W^2; X > 0] \\ &+ \mathbb{E}[\{(\sigma + 1)(-X + 2\bar{A}_-) + 6A_-\}W^2; X < 0]. \end{aligned} \tag{13}$$

All terms on the right are nonnegative. In order, they charge $p \ne p_*$, inward motion of the gap endpoints away from their conditional means, and the spread within each sign class. The last two expectations retain the actual locations of the conditional fluctuations.

Proof. Write $M_{j,\pm} = \mathbb{E}X_\pm^j$. The outer-gap bounds proved above give

$$J_o^+ \le M_{3,+} - 2A_+ M_{2,+} + A_+^2 m, \qquad J_o^- \le M_{3,-} - 2A_- M_{2,-} + A_-^2 m.$$

Adding the central term $m(A_- + A_+)^2$ and substituting in D yields

$$\begin{aligned} D \ge{}& (\sigma - 1)M_{3,+} + (\sigma + 1)M_{3,-} + 6A_+ M_{2,+} + 6A_- M_{2,-} \\ &- 6m(A_-^2 + A_- A_+ + A_+^2). \end{aligned}$$

Conditional centering gives the exact identities

$$M_{2,+} = m\bar{A}_+ + \mathbb{E}[W^2; X > 0], \qquad M_{3,+} = m\bar{A}_+^2 + \mathbb{E}[(X + 2\bar{A}_+)W^2; X > 0],$$

and the analogous identities with $-X, \bar{A}_-$ on the negative class. After substituting $A_- = \ell(p - a_0)$ and $A_+ = \ell(q - a_1)$, the deterministic terms are

$$\ell w\{\mathcal{Q}(p) + 6[d(1-d) + a_0 a_1]\};$$

the remaining terms are exactly the two expectations in (13). The gap sums are nonnegative, so monotone convergence justifies summing countably many gaps. Each expectation is integrable under the finite third-moment assumption; endpoint atoms do not affect the Lebesgue integrals. □

Corollary 3.5 (Simultaneous gap inequality)**.** *For this central gap,*

$$\begin{aligned} D \ge{}& \frac{(6 + 2\sigma)(1 - \theta)^{3/2}}{\sqrt{pq}}(p - p_*)^2 + 6m\{(A_- + A_+)(\Delta_- + \Delta_+) + \Delta_- \Delta_+\} \\ &+ 3(\sigma + 1)A_+\theta_+ + 3(\sigma + 3)A_-\theta_-. \end{aligned} \tag{14}$$

Proof. On the positive support $X \geq A_+$ and $\bar{A}_+ \geq A_+$, so the positive kernel in (13) is at least $3(\sigma+1)A_+$. On the negative support $-X \geq A_-$ and $\bar{A}_- \geq A_-$, the corresponding kernel is at least $3(\sigma+3)A_-$. Finally $m = pq\ell$ and $A_- + A_+ = \ell(1-d)$ identify the two deterministic terms with those in (14). □

4 Conservation and control of the defect

The equality case in Section 3 identifies the extremal Bernoulli law. To connect that geometry to the excess, we integrate the nonnegative contact difference $\Delta = Q - h$ against the zero-bias density. This integral has two useful descriptions. Since Δ vanishes on the support, integration over the gaps produces J and a nonnegative leakage term. The identity $\int \tau g' = \mathbb{E}[Xg(X)]$ gives a second description in terms of moments and the sum distribution. Equating them yields conservation. The coefficient $C_{\mathrm{E}} = (3+\sqrt{10})\phi(0)/6$ then combines the third-moment penalty with the gap term into the nonnegative defect D.

Retain the definitions of a, Q, h and $\Delta = Q - h$ from Section 2. For now let (P, H) be an attained pair at any $c > 0$, so the law has compact support K and $\Delta = 0$ on K.

4.1 Zero-bias integration and the conservation identity

Put $\nu = \mathcal{L}(H - S_{n-1})$ and $\tau(t) = \mathbb{E}[X\mathbf{1}_{X \geq t}]$. Then $\tau \geq 0$, $\int \tau = 1$. Over the bounded complementary intervals (u, v) of the closed support K, define

$$J = \sum \tau_{u,v}(v-u)^2, \qquad J_1 = \sum \tau_{u,v} \int_u^v (t-u)t\,dt,$$
$$J_2 = \sum \tau_{u,v} \int_u^v (t-u)t|t|\,dt, \qquad L = \sum \tau_{u,v} \int_{[u,v)} (t-u)\,d\nu(t).$$

Here $\tau_{u,v}$ is the constant interior value; J and τ agree with the definitions in Section 3. The leakage L is a weighted contribution of the law of $H - S_{n-1}$ inside support gaps, and is nonnegative because its measure and integrand are nonnegative. This sign needs no estimate for C_{n-1}. The integral uses $[u, v)$: the right endpoint belongs to $h(v)$, whereas an atom at the left endpoint has zero weight. If $K \subset [-R, R]$, then $J \leq 2R$, $|J_1| \leq RJ/2$, $|J_2| \leq R^2 J/2$, and $0 \leq L \leq 2R$, so all gap sums converge.

The remainder has four parts: a constant moment and gap term, a linear threshold term, a quadratic threshold term, and the Fourier error E_n. Specifically, let $\phi_z = \phi(H/\sqrt{n})$, $M_4 = \mathbb{E}X^4$, $M_5 = \mathbb{E}|X|^5$, and put

$$K_q = M_5/4 - \rho M_4/2 - 3\beta\gamma/2 + \rho/2, \qquad K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2,$$

$$\Omega = J_1 + (M_4 - 3)/6, \quad \varepsilon = \mathbb{E}q_z(S_n/\sqrt{n}) - \mathbb{E}q_z(Z_0) - 2\gamma\phi_z/(3\sqrt{n}), \quad q_z(x) = (x-z)|x-z|,$$

where $z = H/\sqrt{n}$ and Z_0 is standard normal. Define $E_n = -n^{3/2}\varepsilon/2$ and

$$\mathcal{R}_c = c(3\beta J/2 + 3\rho J_1 - 3J_2 - K_q) + H(cK_1 + \phi_z\Omega) - H^2(c\rho + \gamma\phi_z/2) + E_n. \tag{15}$$

Theorem 4.1 (Conservation at contact)**.** *At an attained pair with $c = C_{\mathrm{E}}$, put $e = \exp(-H^2/(2n))$. Then*

$$\mathcal{R}_c = (n + H^2)a + nc\rho(1-e) + \frac{n\phi(0)e}{6}D + n^{3/2}L. \tag{16}$$

More generally, at an attained pair for any $c > 0$,

$$(n + H^2)a + nc\rho - \frac{n\phi_z}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c. \tag{17}$$

The terms in (16) have separate roles:

Term	Role
$(n+H^2)a$	The excess, weighted by $n+H^2$.
$nc\rho(1-e)$	The loss from moving the normal threshold away from zero.
$n\phi(0)eD/6$	The moment and gap defect from Section 3.
$n^{3/2}L$	A weighted contribution inside the support gaps.
$\mathcal{R}_c$	The moment, threshold and Fourier terms to be bounded.

The four terms on the right of (16) are nonnegative when $a \geq 0$. Solving for the excess makes the role of the remainder explicit:

$$a = \frac{\mathcal{R}_c - nC_{\mathrm{E}}\rho(1-e) - n\phi(0)eD/6 - n^{3/2}L}{n+H^2}. \tag{18}$$

In particular, fix $T \geq 0$, $H_0 \geq 0$ and $N \geq 2$. For any such pair with $a \geq 0$, $n \geq N$, $\mathcal{R}_c \leq T$ and $|H| \leq H_0$, one has

$$D \leq \frac{6T\exp(H_0^2/(2N))}{n\phi(0)}.$$

Theorem 5.1 supplies such numerical bounds under its stated hypotheses. Thus the $1/n$ control of the defect comes from uniform quantitative estimates, rather than from a fixed-law limit.

Proof. 1. Integration over the gaps. Let $\mathcal{A} = \int \tau(Q-h)$. Since $h(t) = \nu([t,\infty))$, one has $h(t) = h(v) + \nu([t,v))$ inside a gap. Thus bounded Fubini gives

$$\int_u^v \Delta(t)dt = (v-u)\Delta(v) + \int_u^v (t-u)(-Q'(t))dt - \int_{[u,v)} (t-u)d\nu(t).$$

An atom at v belongs to $h(v)$ and is not in the last integral; an atom at u is included but multiplied by zero. Because $\Delta = 0$ on the entire closed support, the endpoint terms and the integral over K vanish. Summing gives

$$\mathcal{A} = \frac{\phi_z J}{2\sqrt{n}} + \frac{3c\beta J/2 + (H\phi_z + 3c\rho)J_1 - 3cJ_2}{n^{3/2}} - L. \tag{19}$$

2. Zero-bias integration. Signwise Fubini proves $\int \tau g' = \mathbb{E}[Xg(X)]$ for absolutely continuous g with bounded derivative on the relevant compact range. In particular

$$\int t\tau = \gamma/2, \quad \int t^2\tau = M_4/3, \quad \int |t|^3\tau = M_5/4.$$

Apply that same identity conditionally to $g_s(x) = \min(s+x, H)$ and then use exchangeability. The choice of derivative at its single corner is immaterial under the density τ. It follows directly that

$$\int \tau(t)h(t)dt = \mathbb{E}[X_n \min(S_n, H)] = \frac{1}{n}\mathbb{E}[S_n \min(S_n, H)] = T,$$

where $T = \mathbb{E}[(Z^2 - zZ)\mathbf{1}_{Z\leq z}]$, $Z = S_n/\sqrt{n}$. The last expression vanishes at an endpoint atom. Integrating (5) against τ therefore yields

$$\mathcal{A} = F - \frac{\gamma\phi_z}{2\sqrt{n}} + \frac{H\phi_z(3-M_4)/6 + cK_q}{n^{3/2}} - T. \tag{20}$$

3. The first-moment identity. Set $M = \mathbb{E}[Z\mathbf{1}_{Z\le z}]$. Since $\mathbb{E}[Xh(X)] = M/\sqrt{n}$, integration of $XQ(X)$ against P gives

$$M + \phi_z = -\frac{\gamma H \phi_z}{2n} + \frac{cK_1}{n}. \tag{21}$$

4. Gaussian comparison and assembly. The pointwise identity $q_z(x) = (x-z)^2 - 2(x-z)^2\mathbf{1}_{x\le z}$, with the same identity for a normal variable, gives

$$T - \Phi(z) = -\varepsilon/2 - \gamma\phi_z/(3\sqrt{n}) + z(M + \phi_z) - z^2(F - \Phi(z)).$$

Inserting (21) and $\sqrt{n}(F - \Phi(z)) = c\rho + a$ into (20) gives

$$\begin{aligned}\sqrt{n}\mathcal{A} &= c\rho + a - \gamma\phi_z/6 + P_0/n + \sqrt{n}\varepsilon/2,\\ P_0 &= H^2(c\rho + a + \gamma\phi_z/2) - H\{cK_1 + \phi_z(M_4 - 3)/6\} + cK_q.\end{aligned}$$

Equate this with (19) multiplied by $\sqrt{n}$, then multiply by n and collect terms. This is (17). Substituting $c = C_{\mathrm{E}} = (3 + \sqrt{10})\phi(0)/6$, $\phi_z = \phi(0)e$, and $\gamma + 3J = (3 + \sqrt{10})\rho - D$ proves (16). □

The compensated Fourier representation of this same E_n is proved in Appendix A. Its comparator has transform $e^{-nt^2/2}(1 - in\gamma t^3/6)$. Appendix C retains the stronger identity for arbitrary compactly supported laws, including the residuals that disappear here by contact.

4.2 Why the remainder bound must come first

By Lemma 3.1,

$$D = 3(\rho + 2m - J) + (\sqrt{10}\rho - \gamma - 6m) \ge 0. \tag{22}$$

Specialize from now on to $c = C_{\mathrm{E}}$ and $a \ge 0$. The sharper estimate below keeps the threshold loss. Its coefficient is not known positive until a bound on $\mathcal{R}_c$ has first been established.

Corollary 4.2 (From a remainder bound to a defect bound)**.** *At such a contact pair with $a \ge 0$, assume $n \ge N \ge 2$, $\rho \ge l > 0$, $0 < c_- \le c$, and a proved remainder bound $\mathcal{R}_c \le T < Nc_-l$. A negative T excludes the pair. Otherwise*

$$\begin{gathered}\gamma + 3J > 0, \qquad e \ge 1 - T/(Nc_-l) > 0,\\ \mathcal{U} := (n + H^2)a + n\phi(0)D/6 + n^{3/2}L \le \mathcal{R}_c,\\ \mathcal{U} \le \mathcal{R}_c - \tfrac{1}{2}(c_-l - T/N)H^2.\end{gathered}$$

No condition $D < 2$ or own-score upper bound is required.

Proof. Set $M_c = c\rho - \phi(0)D/6 = \phi(0)(\gamma + 3J)/6$. The contact identity gives

$$eM_c = (1 + H^2/n)a + c\rho + \sqrt{n}L - \mathcal{R}_c/n \ge c\rho - T/n > 0.$$

Thus $M_c > 0$ and, since $D \ge 0$, $M_c \le c\rho$. The density bound follows. Regrouping (16) gives $\mathcal{R}_c = \mathcal{U} + n(1-e)M_c$, proving the unweighted bound. Finally, put $w = H^2/(2n)$, so $e = e^{-w}$. Then

$$n(1-e)M_c = n(e^w - 1)eM_c \ge \frac{H^2}{2}(c\rho - T/n) \ge \frac{H^2}{2}(c_-l - T/N).$$

This is a bound on $\mathcal{U}$, not an improved bound on the raw remainder. □

The corollary is used in two steps. First, an upper bound for $\mathcal{R}_c$ proves $M_c > 0$ and a positive lower bound for e. Second, the now nonnegative term $n(1-e)M_c$ can be subtracted to improve the bound for $\mathcal{U}$ by a multiple of H^2. This is the improvement that helps bound the threshold. It is a bound on $\mathcal{U}$, so it cannot be substituted for the initial upper bound on $\mathcal{R}_c$. Every inequality holds separately at each n, H; common bounds on a closed moment cell therefore also apply when the maximizing law varies with n.

5 Quantitative localization

The zero-defect law is now known. We need a quantitative statement strong enough to control a possible extremizer. Conservation reduces this to an upper bound for its remainder. Contact bounds the support, which permits higher-moment and Fourier estimates. Iterating these estimates gives the following theorem; its two principal outputs for the next section are $nD \le 32.20665378$ and $|H| \le 1.83605680$.

Theorem 5.1 (Fixed-target coarse localization)**.** *Let $n \ge 42$ be an actual index and let an attained fixed-target pair at C_{E} have $a \ge 0$. Assume $C_{n-1} \le C_{\mathrm{E}}$, the external structural inequality* (50)*, and Conditions D.3 and D.5. Then the complete probability cover specified in Appendix D applies to that pair, and simultaneously*

$$\begin{gathered}\mathcal{R}_c \le 2.1995174, \qquad nD \le 32.20665378, \qquad D \le 0.76682509,\\ \rho \le 71527/48000, \qquad |\beta| \le 0.50549963, \qquad |H| \le 1.83605680,\\ \operatorname{supp} P \subset (-157/50, 157/50).\end{gathered}$$

The full analytical proof and all finite conditions are in Appendix D. The Fourier bounds in Appendix A apply uniformly to every actual $n \ge 42$, even when the law and threshold vary with n. Evaluating their scalar majorants at the floor $N = 42$ is therefore sufficient; it does not set the actual sample size equal to 42.

5.1 How the predecessor bound controls support

Lemma 5.2 (Contact bound from the predecessor inequality)**.** *Let (P, H) maximize the excess at $c = C_{\mathrm{E}}$, with $n \ge 42$, $a \ge 0$ and $C_{n-1} \le c$. Put $F_v^{\mathrm{G}}(x) = \Phi(x/\sqrt{v})$ and $\eta_n = n(\sqrt{n/(n-1)} - 1) - 1/2$. Define*

$$\frac{H_n(H,t)}{n^{3/2}} = F_{n-1}^{\mathrm{G}}(H-t) - F_n^{\mathrm{G}}(H) + t(F_n^{\mathrm{G}})'(H) - \frac{t^2-1}{2}(F_n^{\mathrm{G}})''(H).$$

Every support point satisfies

$$0 \le H_n(H,t) + c\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho\eta_n\} - na.$$

Proof. At a support point the contact expression (4) equals zero. Substitute

$$\mathbb{P}(S_{n-1} \le H - t) \le F_{n-1}^{\mathrm{G}}(H-t) + c\rho/\sqrt{n-1}$$

and $F = F_n^{\mathrm{G}}(H) + (c\rho + a)/\sqrt{n}$ into that equality. Collecting the Gaussian terms gives the displayed H_n; the remaining terms give the asserted inequality. The estimate uses the closed distribution function at the same threshold $H - t$. □

A support point is impossible when the right side is negative. The Gaussian expansion and radial derivative estimates in the proof of Theorem D.1 make this exclusion effective on entire exterior rays. This is how the predecessor assumption enters both coarse localization and Proposition 6.2.

5.2 How the bounds improve

The steps have distinct purposes:

1. The structural inequality bounds ρ. Contact and the sharp predecessor then bound the entire support, uniformly in H.

2. Bounded support supplies higher-moment bounds and an event for $X - X'$ under two independent copies of the original law. This event controls the nonzero-frequency peaks in the Fourier integral.

3. Two coarse passes bound the remainder and then the defect, proving that zero is outside the support. Thus the original law has an actual central gap.

4. Four signed-moment passes retain the gap geometry and the signs of the moment terms. They improve the same-law bounds and give the outputs of the theorem.

Choose rational constants $0 < c_- < C_{\mathrm{E}} < c_+$ and $0 < \phi_l < \phi(0)$; their explicit values and proofs are given in Appendices D and B. For example, suppose the coefficient bounds on a cell with $\rho \geq l$ give, for $v = c_-, c_+$,

$$\mathcal{R}_c \leq \max_v \{B_v + F_v|H| - Q_v H^2\}, \qquad Q_v > 0.$$

Completing the square bounds this by $\max_v(B_v + F_v^2/(4Q_v))$. An outward rational upper bound T with $T < Nc_-l$ first establishes the positivity needed in Corollary 4.2. Only then do the larger curvatures $Q_v + (c_-l - T/N)/2$ give a bound U for $\mathcal{U}$, and hence $nD \leq 6U/\phi_l$. This explains both the update and its order: U bounds a nonnegative expression, not the original remainder.

5.3 Why the finite verification covers all witnesses

Proposition 5.3 (Soundness of the finite localization)**.** *Suppose the enclosure and exclusion rules proved in Appendices A and D are evaluated in the stated directions, every required inequality passes, and every closed-cover incidence is checked. Then the resulting bounds hold simultaneously for every attained nonnegative-excess pair in the initial domain at every actual $n \geq 42$ with the sharp predecessor.*

Proof. The initial 24 closed ρ-cells cover the moment range supplied by Theorem D.1. Fix one possible witness in one of these cells. Every subdivision covers its parent. Intersecting necessary moment inequalities keeps this witness; the original β-interval is restored separately for each γ-slice. A child is discarded only by an incompatible necessary inequality. In particular, a positive numerator with zero denominator is treated using the original moment matrix, and a zero-over-zero correction is dropped.

For each surviving child, a support bound is proved before higher moments are used. The actual-event and Fourier estimates then bound the remainder, and the ordered update bounds the nonnegative expression in conservation. Taking maxima of upper bounds and minima of lower bounds can enlarge the possible set but cannot remove the fixed witness. A parent disappears only if all descendants have been excluded. Induction over the two coarse and four signed passes preserves this invariant.

The final closed probability panels cover the surviving probability intervals. At a shared endpoint the cell contributes to every incident panel. Pooling the simultaneous bounds therefore retains the conclusions for the same law. Finally, the analytical all-index comparisons in the Fourier appendix justify each floor evaluation at every actual $n \geq 42$. The numbers of processed boxes are accounting checks, not substitutes for this coverage argument. □

6 From small defect to a common support region

The localization theorem bounds nD and $|H|$. Lemma 3.4 bounds D below by nonnegative costs, so a small defect makes each cost small: the sign probability must be close to p_*, the

support gap must nearly reach the conditional means, and the spread within each sign class must be small. Proposition 6.1 makes these conclusions quantitative using the defect alone. Proposition 6.2 then uses contact and the sharp predecessor bound to control the entire conditional support. Together with the threshold bound, these conclusions give the domain needed by Theorem 7.1. Throughout this section set

$$\sigma = \sqrt{10}, \qquad p_* = (4-\sigma)/2, \qquad B_* = 32.20665378.$$

The law, support gaps and sign split are those of Section 3; in particular D is the actual gap defect, not a moment surrogate. Write the central gap, once acquired, as $(-A_-, A_+)$, using the notation of Section 3. The first use of $n \geq 700$ is the inequality $B_*/700 < .047$ below; the count comparison also uses this sample-size bound in its error estimates.

Proposition 6.1 (Direct geometry)**.** *Let X be compactly supported, centered and of variance one. If $n \geq 700$ and $nD \leq B_*$, its closed support has a gap $(-A_-, A_+)$ across zero, with $A_-, A_+ > 0$. In the actual decomposition $X = \ell(B_0 + U - p)$, put*

$$a_0 = p - A_-/\ell, \quad a_1 = q - A_+/\ell, \quad d = a_0 + a_1,$$
$$\theta = \mathbb{E}\{X - \mathbb{E}(X \mid B_0)\}^2, \qquad \lambda = n\mathbb{E}U^2 = n\theta/\ell^2.$$

Then $a_0, a_1 \geq 0$, $\mathbb{E}(U \mid B_0) = 0$, $U_0 \leq a_0$, $U_1 \geq -a_1$, and

$$\frac{3}{8} < p < \frac{93}{200}, \qquad \theta < .0036, \qquad d < .004, \qquad \ell > 1.996, \tag{23}$$
$$\rho < \frac{9}{8}, \qquad .066148 < \beta < .2527, \qquad \lambda < \frac{7}{20}. \tag{24}$$

These conclusions require no contact or predecessor hypothesis.

The estimates are obtained in the following order. Each preliminary bound supplies a hypothesis for a sharper one later in the proof.

Input	Purpose of the output
$D < .047$ and the zero-in-support bound	Establish a central gap and nonnegative inward displacements.
The joint defect budget	Obtain $.36 < p < .48$, $\theta < .006$ and $\ell > 1.99$ before taking derivatives of the refined bounds.
The inward-motion term	Give $d < .004$.
Monotonicity in d and θ	Give the final probability interval, $\theta < .0036$ and $\ell > 1.996$.
The conditional cubic costs	Control the total count variance λ, rather than only the variance of X.
Contact and the predecessor bound	Exclude the two outer support rays in Proposition 6.2.

Proof. 1. A central gap and preliminary bounds. We have $D < D_0 := .047$, because $700D_0 > B_*$. If zero were in the closed support, Lemma 3.1 would give $D \geq (\sigma-1)\rho \geq \sigma - 1 > D_0$. Thus the central gap exists; centering supplies both signs. Conditional means imply $a_j \geq 0$ and $0 \leq d < 1$. Put $v = pq$ and $w = 1 - \theta = v\ell^2$, and use the probability polynomial $\mathcal{Q}$ from Section 3. We first retain all the costs of the actual conditional noises. Let

$$\int f\, d\mu_\pm = \mathbb{E}\big[(X - \mathbb{E}(X \mid B_0))^2 \mathbf{1}_{\{\pm X > 0\}} f(|X|)\big].$$

Their masses sum to θ. In this notation the joint budget (13) reads

$$D \geq \ell w \mathcal{Q} + E_0 + \int k_+\, d\mu_+ + \int k_-\, d\mu_-, \tag{25}$$

where

$$E_0 = 6\ell w\{d(1-d) + a_0 a_1\},$$
$$k_+(x) = (\sigma-1)(x+2\ell q) + 6A_+, \qquad k_-(x) = (\sigma+1)(x+2\ell p) + 6A_-.$$

The measures $\mu_\pm$ abbreviate the displayed variance-weighted expectations under the same marginal P.

Hereafter use $3.162277660 < \sigma < 3.162277661$. The weaker gap estimate $J \le \rho + 2mA_-A_+$, together with $\rho \ge 1$, $\gamma \le \rho$ and $m \le 1/2$, gives

$$A_-A_+ \ge z := \frac{3.162277660 - 1 - D_0}{3} > 0, \qquad w \ge A_-A_+ \ge z.$$

The second inequality follows from $A_- \le \ell p$, $A_+ \le \ell q$. All costs in (25) are nonnegative, so

$$D \ge \frac{w^{3/2}\mathcal{Q}}{\sqrt{v}} \ge 2z^{3/2}\mathcal{Q}.$$

Let $\mathcal{Q}_-$ replace σ in its polynomial expression by the lower rational bound. The two exact tests

$$4z^3\mathcal{Q}_-(.36)^2 > D_0^2, \qquad 4z^3\mathcal{Q}_-(.48)^2 > D_0^2$$

exclude $p \le .36$ and $p \ge .48$, respectively: the squared distance to $p_* \in (.36, .48)$ is monotone outside that interval.

2. Preliminary noise and inward-displacement bounds. On $.36 < p < .48$, $\ell \ge 2\sqrt{z} > 1.67$, and the noise coefficients satisfy

$$k_+(x)/\ell \ge q\{(\sigma+5)z/w + 2(\sigma-1)\} > 5.2,$$
$$k_-(x)/\ell \ge p\{(\sigma+7)z/w + 2(\sigma+1)\} > 5.2.$$

For the first use $x \ge A_+$ and $A_+ \ge z/(\ell p)$; for the second use their negative counterparts. Since $w \le 1$, the constants follow by taking $q = .52$ and $p = .36$. Therefore $D \ge 1.67(5.2)\theta$, whence $\theta < .006$, $w > .994$ and $\ell > 1.99$. Also $A_- + A_+ \ge 2\sqrt{z}$ gives

$$d \le 1 - 2\sqrt{zv/w} < .2, \qquad v \ge .36(.64).$$

Evaluating the kernels at their lower support endpoints in (25) now yields

$$D \ge \ell\{w\mathcal{Q} + 6wd(1-d) + \theta\min(C_+ - b_+d, C_- - b_-d)\}, \tag{26}$$

where

$$C_+ = 3(\sigma+1)q, \quad C_- = 3(\sigma+3)p, \quad b_+ = \sigma+5, \quad b_- = \sigma+7.$$

Each branch increases in d on the current domain, since $6(.994)(.6) - .006(3.162277661+7) > 0$; both noise prices are positive. The inward cost alone excludes $d \ge .004$, because

$$1.99(.994)6(.004)(.996) > .047.$$

3. The final probability and variance bounds. After replacing d by zero, the derivative in θ of either branch $\sqrt{(1-\theta)/v}\{(1-\theta)\mathcal{Q} + \theta C\}$ has the sign of $(2-3\theta)C - 3(1-\theta)\mathcal{Q}$. It is positive: $C > 6.49$, $\mathcal{Q} < .05$ and $\theta < .006$. Hence $D \ge \mathcal{Q}/\sqrt{v}$. Since $\sqrt{v} \le 1/2$, it follows that $D \ge 2\mathcal{Q}(p)$. Moreover

$$\frac{3}{8} < p_* < \frac{21}{50} = \frac{1}{2}\left(\frac{3}{8} + \frac{93}{200}\right).$$

Every p outside $(3/8, 93/200)$ is therefore at least as far from p_* as $3/8$, so $\mathcal{Q}(p) \geq \mathcal{Q}(3/8)$. The single rational comparison

$$\mathcal{Q}_-(3/8) = \frac{37936011}{1600000000} > \frac{47}{2000} = \frac{D_0}{2}$$

contradicts $D < D_0$ on both outside rays, including the endpoints. Thus $3/8 < p < 93/200$. Since

$$C \geq \min\{3(\sigma+1)(107/200),\ 3(\sigma+3)(3/8)\},$$

the bound $D \geq \ell\theta C$, with $\ell > 1.99$, improves $\theta < .0036$ and consequently $\ell > 1.996$. Here the d-dependent terms may be dropped together, since $6w(1-d) - \theta b_\pm > 0$ on the current domain. This keeps the displacement cost together with its signed noise correction.

4. The remaining moment bounds. The exact conditional cubic expansion gives

$$\rho = w^{3/2}\rho_B(p) + \int (x + 2\ell q)\, d\mu_+ + \int (x + 2\ell p)\, d\mu_-, \qquad \rho_B(p) = \frac{1-2v}{\sqrt{v}}.$$

Because the two integrands are priced by at least $\sigma - 1$ in (25),

$$\rho \leq \rho_B(p) + D/(\sigma-1) < 1.1 + .047/(54/25) < 9/8.$$

Here $v \geq 15/64$ implies $\rho_B < 1.1$ by squaring positive quantities. The signed-second-moment identity gives

$$1 - 2p - 2q\theta \leq \beta \leq 1 - 2p + 2p\theta;$$

its endpoints imply $.066148 < \beta < .2527$.

5. From conditional spread to total count variance. A bound on θ alone would not control $\lambda = n\theta/\ell^2$ uniformly in n. We therefore retain more of the cubic cost. A one-sided moment inequality supplies a variance-squared term divided by the inward displacement; pairing it with the displacement cost by AM–GM will give a larger linear coefficient of λ. If Z is centered, $Z \leq a$, $a > 0$, and $t = \mathbb{E}Z^2$, then

$$0 \leq \mathbb{E}[(a - Z)(Z + t/a)^2] = at - \mathbb{E}Z^3 - t^2/a.$$

Apply this to U_0 and $-U_1$. A zero endpoint forces the corresponding noise to vanish, so no division by zero is needed. For $\lambda_0 = nq\mathbb{E}U_0^2$, $\lambda_1 = np\mathbb{E}U_1^2$, the same gap-budget expansion, with these two cubic bounds inserted, gives

$$\begin{aligned} nD/\ell^3 \geq\ & nv\mathcal{Q} + 6nv\{d(1-d) + a_0a_1\} \\ & + \{3(\sigma+3)p - (\sigma+7)a_0\}\lambda_0 + \{3(\sigma+1)q - (\sigma+5)a_1\}\lambda_1 \\ & + \frac{(\sigma+1)\lambda_0^2}{nqa_0} + \frac{(\sigma-1)\lambda_1^2}{npa_1}. \end{aligned}$$

Terms with $a_j = 0$ are absent. Set $\bar{d} = .004$ and use $d(1-d) + a_0a_1 \geq (1-\bar{d})(a_0 + a_1)$. The reciprocal term is essential: pairing its variance-squared cost with the inward-displacement cost gives a linear cost in λ. Specifically, AM–GM gives

$$6nv(1-\bar{d})a_0 + \frac{(\sigma+1)\lambda_0^2}{nqa_0} \geq 2\lambda_0\sqrt{6(1-\bar{d})(\sigma+1)p}.$$

Thus

$$nD \geq \ell^3\{nv\mathcal{Q} + \lambda_0 Y_0(p) + \lambda_1 Y_1(p)\}, \tag{27}$$

where

$$Y_0 = 3(\sigma+3)p - (\sigma+7)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma+1)p},$$
$$Y_1 = 3(\sigma+1)q - (\sigma+5)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma-1)q}.$$

Each increases in its respective argument p or q. At $p = 3/8$ and $q = 107/200$, lower bounds 3 and 2.6 for the roots give $Y_0 > 12.8$, $Y_1 > 11.8$. Since $\lambda_0 + \lambda_1 = \lambda$,

$$B_* \geq nD > 1.996^3(11.8)\lambda \quad (\lambda > 0), \qquad 1.996^3(11.8)(.35) > B_*.$$

This proves $\lambda < .35$; at zero noise it is immediate. □

Proposition 6.2 (Two contact rays)**.** *Under the hypotheses of Proposition 6.1, suppose in addition that $C_{n-1} \leq C_{\mathrm{E}}$ and that (P, H) maximizes the excess at C_{E}, with nonnegative maximum. Then*

$$-.6 < U_0 \leq a_0, \qquad -a_1 \leq U_1 < .6.$$

The support conclusion is uniform in the raw threshold H.

Proof. A support point too far from its conditional mean would make the contact upper expression (52) negative. The all-threshold Gaussian heat estimate and its radial derivative bound from the proof of Theorem D.1 give, for $t > \sqrt{3}$,

$$H_n(H, \pm t) \leq \overline{H}(t) := S^3 \left\{ \frac{2}{5} \left(\frac{t^3 - 3t}{6} + \frac{|t^4 - 6t^2 + 3|}{72} \right) + \frac{t^5}{4100} + \frac{3t}{820} + \frac{17(3t^2-2)}{78720} \right\},$$
$$|\partial_t H_n(H, \pm t)| < g_{\mathrm{rad}}(t) := \frac{2}{5}(t^2/2 + 1/2 + \eta), \qquad S = 81/80, \quad \eta = 8/875.$$

The first bound follows directly from (53): $\|\phi'''\|_\infty < 2\phi(0)$, $n - 1 \geq 41$, and $\sqrt{n-1} > 6$. No derivative of the absolute-value majorant is taken.

At a support point of magnitude t, contact requires a nonnegative upper expression $H_n(H, \pm t) + C_{\mathrm{E}} Q_b(t)$, where

$$Q_b(t) = -t^3 + \tfrac{3}{2} r t^2 + 3bt + \eta r, \quad r = 9/8, \quad b = \begin{cases} -.066148 & \text{negative ray,} \\ .2527 & \text{positive ray.} \end{cases}$$

These replacements follow with their signs from (24); the discarded excess term is nonpositive. Take $c_0 = 819/2000 < C_{\mathrm{E}}$ and the two fixed boundaries

$$t_- = 1.996(3/8 + .6) = 1.9461, \qquad t_+ = 1.996(107/200 + .6) = 2.26546.$$

For a fixed raw threshold H, abbreviate the appropriate radial function $H_n(H, \pm t)$ by $H_n(t)$, with derivative taken in the positive magnitude t. Set

$$\Xi(t) = g_{\mathrm{rad}}(t) + c_0 Q_b'(t).$$

Exact rational substitution gives the following conservative bounds:

Boundary quantity	t_-, negative ray	t_+, positive ray
$\overline{H}(t) + c_0 Q_b(t)$	$< -2/5$	$< -13/100$
$\Xi(t)$	< -1	$< -8/5$
$\Xi'(t)$	$< -13/5$	$< -16/5$

Both boundaries satisfy $t^2 > 3$. The constant second derivative is

$$\Xi'' = \frac{2}{5} - 6c_0 = -\frac{2057}{1000} < 0.$$

Consequently Ξ' and then Ξ remain negative on each outward ray. Because $g_{\mathrm{rad}} > 0$, the identity $Q_b' = (\Xi - g_{\mathrm{rad}})/c_0$ gives $Q_b' < 0$ there. At the boundary $\overline{H} \geq 0$, so the first table entry gives $Q_b < 0$; this sign also persists outward. Since $C_{\mathrm{E}} > c_0$, we obtain

$$H_n(t_\pm) + C_{\mathrm{E}} Q_b(t_\pm) < 0, \qquad H_n'(t) + C_{\mathrm{E}} Q_b'(t) \leq g_{\mathrm{rad}}(t) + c_0 Q_b'(t) = \Xi(t) < 0.$$

The contact upper expression therefore cannot be nonnegative anywhere on either ray. This excludes every support point there, including boundary atoms. We differentiate H_n itself; the absolute-value majorant $\overline{H}$ is used only at the two boundary points.

On the negative class, $X > -t_-$ and $\ell > 1.996$ give $U_0 = X/\ell + p > -(3/8 + .6) + p > -.6$. On the positive class, $X < t_+$ gives $U_1 = X/\ell + p - 1 < 107/200 + .6 + p - 1 < .6$. The inward endpoints and conditional centering are the actual ones from Proposition 6.1. □

By (6), the law compared in the next section is exactly the original law X, with the same threshold H. The sign probability and the signed moments refer throughout to this chosen law.

7 The direct comparison with a binomial law

In count coordinates, conditioning on the number of positive summands expresses the sum as an integer count plus centered noise. We compare its CDF with the corresponding binomial CDF. Noise can move mass out of a count near the threshold and bring mass in from counts above it. The proof shows that the outgoing loss, together with the normal and third-moment corrections, exceeds all incoming contributions. The quantity $\lambda = n\mathbb{E}U^2$ measures the total noise variance in count units.

Theorem 7.1 (The common-strip comparison)**.** *Let $n \geq 700$ be an integer, $B \sim \mathrm{Bernoulli}(p)$, and $\mathbb{E}(U \mid B) = 0$, where $p \in [3/8, 93/200]$. Write U_j for the conditional noise given $B = j$, and suppose*

$$-3/5 \leq U_0 \leq a_0, \qquad -a_1 \leq U_1 \leq 3/5, \qquad a_0, a_1 \geq 0, \quad a_0 + a_1 \leq 1/250.$$

Put $q = 1 - p$, $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $X = (B + U - p)/\sqrt{v + s}$ and $\rho = \mathbb{E}|X|^3$. For independent copies and every raw threshold $|H| \leq 46/25$,

$$\sqrt{n}\left\{\mathbb{P}\left(\sum_{i=1}^{n} X_i \leq H\right) - \Phi(H/\sqrt{n})\right\} - C_{\mathrm{E}}\rho < -\lambda/100 \qquad (0 < \lambda \leq 7/20). \tag{28}$$

At $\lambda = 0$ the same excess is strictly negative.

To prove the theorem, condition on $K = \sum_{i=1}^{n} B_i$, which has law $\mathrm{Bin}(n, p)$. Given $K = j$, the count-unit noise W_j is the sum of j independent copies of U_1 and $n - j$ independent copies of U_0. The threshold in these units is

$$t = np + H\sqrt{v + s}, \qquad \left\{\sum X_i \leq H\right\} = \{K + W_K \leq t\}.$$

We compare this probability with the binomial CDF and then account for the changes in the normal term and the third absolute moment. A count below t can leave the event, while

a count above t can enter. The aim is to show that the loss and the favorable corrections exceed the total gain from all counts above the threshold.

For $0 < \lambda \le 1/100$, the argument keeps both the loss and every error proportional to λ. It uses one nearby count and a fourth-moment bound for all entering counts. For $1/100 \le \lambda \le 7/20$, a finite summation identity separates the crossing loss from errors in interpolating the binomial CDF. A periodic sine function then gives a loss from five counts that is uniform in the phase $t - \lfloor t \rfloor$. Its interpolation error has an upper bound of order $1/n$ that does not vanish with λ, which explains the separate small-noise argument. The two ranges share their endpoint. At $\lambda = 0$ the law is exactly Bernoulli. These comparisons use the explicit hypotheses of this theorem; the preceding sections obtain those hypotheses for a maximizer.

7.1 Three elementary ingredients

Write $b_j = \binom{n}{j} p^j q^{n-j}$, $F_B(j) = \sum_{i \le j} b_i$, and $\rho_B = (1-2v)/\sqrt{v}$. We use Schulz's strict finite-binomial theorem [11, Theorem 1] in the form

$$\sqrt{n}\{F_B(j) - \Phi((j-np)/\sqrt{nv})\} < C_{\mathrm{E}}\rho_B \quad (0 < p < 1, n < \infty). \tag{29}$$

Its strictness concerns every finite binomial law; it is not inferred from the limiting lower approximation for the universal constant.

The factorial estimates used throughout are finite Stirling bounds:

$$\log(j!) = (j + \tfrac{1}{2})\log j - j + \tfrac{1}{2}\log(2\pi) + r_j, \qquad 0 < r_j < 1/(12j).$$

For completeness, with $x = 1/(2j+1)$, $r_j - r_{j+1} = \operatorname{arctanh}(x)/x - 1 = \sum_{l \ge 1} x^{2l}/(2l+1)$ lies strictly between 0 and $1/(12j(j+1))$. Wallis's product fixes the limiting constant, so $r_j \to 0$; summing these differences proves the bounds. One useful consequence is

$$\max_j \mathbb{P}\{\mathrm{Bin}(M,p) = j\} < \frac{41}{100\sqrt{Mv}} \quad (M \ge 43, p \in [1/3, 2/3]). \tag{30}$$

Indeed a mode has $z = j/M = p + \delta/M$, $|\delta| \le 2/3$. Since $v \ge 2/9$ and $|1-2p| \le 1/3$, $z(1-z) \ge v(1 - 1/M - 2/M^2) > 39v/40$. Stirling, the nonnegative relative entropy, and $e^{1/516} < 516/515$ bound the mass by $(2/5)\sqrt{40/39}(516/515)/\sqrt{Mv}$, which is smaller than the right side of (30) by a rational square comparison.

Finally the cubic correction has the favorable sign on the full strip. Let $s_j = \mathbb{E}U_j^2$, so $s = qs_0 + ps_1$, and set $r = 3/8$, $d = 1/250$, $v_* = 15/64$. The conditional supports have fixed signs relative to p, since $d < r$. Expansion of the raw third absolute moment gives

$$\mathbb{E}|B + U - p|^3 = v(1-2v) + 3v(s_0 + s_1) - q\mathbb{E}U_0^3 + p\mathbb{E}U_1^3.$$

The inward bounds imply $\mathbb{E}U_0^3 \le a_0 s_0$, $\mathbb{E}U_1^3 \ge -a_1 s_1$. Together with $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$ for $x \ge 0$, this yields

$$\begin{aligned}(v+s)^{3/2}(\rho - \rho_B) &\ge s\left\{3r - d - \frac{3}{2}(1-2v) - \frac{3(1-2v)s}{8v}\right\} \\ &\ge s\left\{\frac{2593}{8000} - \frac{17}{20}s\right\} > 0, \qquad 0 < s \le \frac{1}{2000}.\end{aligned} \tag{31}$$

Here $s \le (7/20)/700 = 1/2000$, and the displayed coefficient is $3r - d - \frac{3}{2}(1-2v_*)$. Thus every nonnegative coefficient of $\rho_B - \rho$ can be discarded in an upper comparison.

7.2 Small noise: one count and the whole incoming tail

Assume $0 < \lambda \le \lambda_s = 1/100$. Choose an integer k with

$$t = np + H\sqrt{v+s} = k + u, \qquad -3/10 \le u \le 7/10, \qquad \delta = k - np.$$

The phase interval has length one, so such an integer always exists. It need not be the floor anchor used in the larger-noise proof. Let W_j have the law of the sum of j independent U_1 variables and $n-j$ independent U_0 variables. For $C \ge 0$, subtract the signed binomial excess at k from the actual excess:

$$\begin{aligned}\Delta_C = \sqrt{n}\left\{\mathbb{P}\left(\sum X_i \le H\right) - \Phi(H/\sqrt{n})\right\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B.\end{aligned}$$

Exact conditioning gives

$$\begin{aligned}\mathbb{P}\left(\sum X_i \le H\right) - F_B(k) = & -\sum_{i=0}^{k} b_{k-i}\mathbb{P}(W_{k-i} > u+i) \\ & + \sum_{i=1}^{n-k} b_{k+i}\mathbb{P}(W_{k+i} \le u-i).\end{aligned}$$

The event is closed: count $k-i$ is lost when $W_{k-i} > u+i$, whereas count $k+i$ enters when $W_{k+i} \le u-i$. Thus an atom at the threshold remains in the event in both sums. Define

$$g_k(u) = \frac{\Phi((\delta+u)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{b_k}, \qquad O = \mathbb{P}(W_k > u) + g_k(u),$$

$$T_{\mathrm{in}} = \sum_{i=1}^{n-k} \mathbb{P}(W_{k+i} \le u-i), \qquad \mathcal{R} = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\} + C(\rho_B - \rho).$$

Keeping the own-count loss and discarding the other nonpositive outgoing terms gives

$$\Delta_C \le -m_k(O - r_{\max}T_{\mathrm{in}}) + \mathcal{R}, \qquad m_k = \sqrt{n}\, b_k, \tag{32}$$

whenever $b_{k+i}/b_k \le r_{\max}$ for all legal incoming counts.

Uniform normalization.

Set $D_* = 5/3$, $b_* = 1/20$, $T_* = 101/10000$ and $c_* = 2/525$. Because $v \ge v_*$ and $v + s < 1/4$, $|t - np| < 23/25$. Consequently $|\delta| < D_*$ globally and $|\delta| < 1$ when $|u| \le b_*$. Write $\mathcal{K}(z,p) = z\log(z/p) + (1-z)\log((1-z)/q)$ for binomial relative entropy; this is distinct from the physical gap defect D. The segment between p and k/n lies inside $[.37, .47]$; in particular $1 \le k \le n-1$. On this segment

$$z(1-z) > 23/100, \quad |\partial_z^3 \mathcal{K}(z,p)| < 5, \quad |(\log(z(1-z)))'| < 6/5, \quad (12z(1-z))^{-1} < 3/8.$$

Stirling and entropy Taylor expansion therefore give

$$m_k = \frac{\phi(0)}{\sqrt{v}} e^{-\delta^2/(2nv)+e}, \qquad n|e| < Z(D_0) := 3D_0/5 + 3/8 + D_0^3/840 \tag{33}$$

for $|\delta| \le D_0 \le D_*$. The three terms respectively bound the logarithmic prefactor, factorial remainder and entropy remainder. Also

$$\log g_k'(z) = -\tfrac{1}{2}\log(1 + s/v) + \frac{\delta^2}{2nv} - \frac{(\delta+z)^2}{2n(v+s)} - e,$$

so, on an integration interval $|z| \le b_0$,

$$n|\log g_k'(z)| \le \frac{11}{5}\lambda + \frac{2D_0 b_0 + b_0^2}{2v_*} + Z(D_0).$$

The coefficient $11/5$ exceeds $1/(2v_*) + D_*^2/(1400v_*^2)$. Using $e^{-x} \ge 1-x$, $e^x \le (1-x)^{-1}$ and $\phi(0) > 199/500$ now gives

	$\lvert u\rvert \le 1/20$	$-3/10 \le u \le 7/10$
m_k lower bound	$3/4$	$3/4$
g_k' lower bound	$99/100$	$49/50$
g_k' upper bound	$101/100$	$51/50$
$r_{\max}$ upper bound	$51/50$	$6/5$

(34)

The derivative estimates hold on the whole segment from 0 to u. For the global ratio, (30) gives $\sqrt{n}\max_j b_j < 17/20$, hence $r_{\max} < (17/20)/(3/4) < 6/5$. In the inner interval the first ratio is

$$\frac{b_{k+1}}{b_k} = 1 + \frac{p-\delta-1}{nv + q(\delta+1)} < 1 + \frac{93/200}{700v_*} < 51/50;$$

every later ratio is at most one, since $p - \delta < 2$. By (31) the moment correction is nonpositive. Differentiating the normal CDF and using $\phi(0)/(2v_*^{3/2}) < 2$ therefore gives

$$\mathcal{R} < 2\lambda/n \quad (|u| \le b_*), \qquad \mathcal{R} < 4\lambda/n \quad \text{on the full phase interval.} \tag{35}$$

Indeed the positive displacement is bounded by 1 or $D_* < 2$.

No truncation of the incoming tail.

Since $|s_1 - s_0| \le (8/3)s$, the actual own-count variance $t_0 = \mathbb{E}W_k^2$ satisfies $.99\lambda < t_0 < 1.01\lambda \le T_*$. For each incoming count,

$$t_i = \mathbb{E}W_{k+i}^2 \le \lambda\{1 + c_*(D_0 + i)\}, \qquad \mathbb{E}W_{k+i}^4 \le A^2 t_i + 3t_i^2, \quad A = 3/5,$$

where $D_0 = 1$ in the inner interval and $D_0 = D_*$ otherwise. The moment bound follows from independence and $|U_j| \le A$. For $a = 1 + c_* D_0$ put

$$K(a) = A^2\left(\frac{13a}{12} + \frac{5c_*}{4}\right) + 3\lambda_s\left(\frac{13a^2}{12} + \frac{5ac_*}{2} + \frac{5c_*^2}{3}\right).$$

Fourth-moment Markov at distance $i - u$, and extension of the nonnegative sum to all positive integers, give

$$\begin{aligned} T_{\text{in}} &< \frac{K(1+c_*)}{(1-b_*)^4}\lambda < .53\lambda \quad (|u| \le b_*), \\ T_{\text{in}} &< \frac{.43\lambda}{(1-u)^4} \qquad\qquad (0 \le u \le .7), \\ T_{\text{in}} &< .43\lambda \qquad\qquad (u \le 0). \end{aligned} \tag{36}$$

Here $K(1 + c_* D_*) < .43$, and $\sum i^{-4} < 13/12$, $\sum i^{-3} < 5/4$, $\sum i^{-2} < 5/3$. These series bounds follow by retaining respectively five, three and five terms and estimating the remaining sum by an integral. For $u \ge 0$ use $i - u \ge (1-u)i$; for $u \le 0$ use $i - u \ge i$.

The loss at the threshold count.

The variance of the count-unit noise already predicts the scale of this loss. Cauchy–Schwarz and Hölder applied to the centered sum W_k give

$$\mathbb{P}(W_k > 0) \geq \frac{(\mathbb{E}|W_k|)^2}{4t_0} \geq \frac{t_0^2}{4\mathbb{E}W_k^4}.$$

Indeed $\mathbb{E}(W_k)_+ = \mathbb{E}|W_k|/2$ and $t_0 \leq (\mathbb{E}|W_k|)^{2/3}(\mathbb{E}W_k^4)^{1/3}$. The fourth-moment bound $\mathbb{E}W_k^4 \leq A^2 t_0 + 3t_0^2$ makes this at least $t_0/[4(A^2+3t_0)] > (16/25)t_0$ on $t_0 \leq T_*$. This zero-threshold estimate explains why a loss proportional to the variance is possible. We use a quartic minorant to obtain a bound that also pays for moving the threshold and for the normal increment.

Lemma 7.2 (Loss at a nearby count threshold)**.** *Under the small-noise hypotheses and for* $|u| \leq b_*$,

$$O > \frac{3}{5}t_0 > .59\lambda. \tag{37}$$

Proof. We first use a moment minorant when $|u| \leq 3t_0/2$, then a one-sided variance bound for the remaining phases. Set $\alpha = 1/2$ and $m_h = 5\alpha^4 = 5/16$. The factored quartic

$$h_\alpha(x) = \frac{-16x^4 + 12x^2 + 4x}{5} = \frac{4}{5}x(1-x)(2x+1)^2 \leq \mathbf{1}_{(0,\infty)}(x) \tag{38}$$

is a global minorant for the strict event. Its factorization makes it nonpositive for $x \leq 0$ and $x \geq 1$. On $(0,1)$ the maximum occurs at $x = (1+\sqrt{3})/4$ and equals $(9+6\sqrt{3})/20 < 1$; the last comparison follows from $6\sqrt{3} < 11$. Thus it may be integrated against the conditional sum W_k. Each summand is bounded by $3/5$, whereas the sum can exceed that bound; the minorant is valid on the entire real line. The expectation of the quartic uses moments through order four, supplied by independence and the bounded summands. Let $r = zt_0$ and set

$$F_{\varepsilon,b}(z,t) = \frac{3\alpha^2 - A^2 - 3t}{m_h} + \left(\frac{2\varepsilon}{5\alpha} + b - \frac{4At}{m_h}\right)z + \frac{3\alpha^2 t - 6t^2}{m_h}z^2 - \frac{t^3 z^4}{m_h}.$$

Expanding $h_\alpha(W_k + r)$ for $u = -r$, or $h_\alpha(W_k - r)$ for $u = r$, and using $|\mathbb{E}W_k^3| \leq At_0$, $\mathbb{E}W_k^4 \leq A^2 t_0 + 3t_0^2$, yields

$$O/t_0 \geq F_{1,-101/100}(z, t_0) \quad (u = -r), \qquad O/t_0 \geq F_{-1,99/100}(z, t_0) \quad (u = r).$$

Both functions exceed $3/5$ on $0 \leq z \leq 3/2$, $0 \leq t \leq T_*$. Indeed $\partial_t^2 F = -(12z^2 + 6tz^4)/m_h \leq 0$, so it suffices to bound the two edges $t = 0, T_*$. Replacing z^4 by $(9/4)z^2$ in its negative term gives a lower quadratic whose coefficient of z^2 is

$$q_{\mathrm{quad}}(t) = \frac{3\alpha^2 t - 6t^2 - (9/4)t^3}{m_h}, \qquad q_{\mathrm{quad}}(0) = 0, \quad q_{\mathrm{quad}}(T_*) = \frac{27842487291}{1250000000000} > 0.$$

For the negative phase, its derivative at the right endpoint is $-21/100$ when $t = 0$, and is less than $-1/5$ when $t = T_*$. The quadratic therefore decreases throughout the interval. For the positive phase, its derivative at zero is $19/100$ or $7027/62500$, so it increases. The resulting endpoint minima satisfy

	$t = 0$	$t = T_*$
$F_{1,-101/100}$ lower bound at $z = 3/2$	$933/1000$	$> 769/1000$
$F_{-1,99/100}$ lower bound at $z = 0$	$156/125$	$3597/3125$

All four exceed 3/5. Concavity in t then covers the entire rectangle.

For negative phases $3t_0/2 \le r \le b_*$, Cantelli gives

$$O \ge \frac{r^2}{t_0 + r^2} - \frac{101}{100} r > \frac{3}{5} t_0.$$

Its derivative at $r = 3t_0/2$ is positive and first increases, then decreases. The minimum is therefore at an endpoint. At the first use $t_0 \le T_*$ after division by t_0; at the other use $b_*^2/(T_* + b_*^2) - (101/100)b_* > (3/5)T_*$. For positive phases beyond this switch, $O \ge (99/100)r \ge (297/200)t_0 > (3/5)t_0$. The switch lies below b_* because $3T_*/2 < b_*$. The value $u = 0$ is included, with $h_\alpha(0) = 0$. □

The outer phases and the final margin.

For $u = -r$, $b_* \le r \le 3/10$, Cantelli gives

$$O \ge \frac{r^2}{T_* + r^2} - \frac{51}{50} r > 7/50 \ge 14\lambda.$$

The derivative is positive at b_* and has at most one downward zero; both endpoint values exceed 7/50. For $b_* \le u \le 7/10$,

$$O - r_{\max} T_{\text{in}} \ge \frac{49}{50} u - \frac{(6/5)(43/100)\lambda_s}{(1-u)^4} > 1/25 \ge 4\lambda.$$

This function is concave and both endpoints exceed 1/25. In the inner interval, (36) and (37) give $O - r_{\max} T_{\text{in}} > (.59 - (51/50).53)\lambda$. Substitute these three margins and (35) into (32), using $m_k > 3/4$. The smallest final margin is the inner one, and

$$\frac{3}{4}\left(\frac{59}{100} - \frac{51}{50}\frac{53}{100}\right) - \frac{2}{700} = \frac{4787}{140000} > \frac{1}{100}.$$

Hence $\Delta_C < -\lambda/100$. Taking $C = C_{\text{E}}$ and applying (29) proves (28) for $0 < \lambda \le 1/100$.

7.3 Larger noise: five counts and two endpoint checks

Now assume $1/100 \le \lambda \le 7/20$ and choose $k = \lfloor t \rfloor$, $u = t - k \in [0, 1)$, $w = t - np$. The following constants will be used only in this subsection:

$$\begin{gathered} n_0 = 700, \quad p_- = 3/8, \quad p_+ = 93/200, \\ r = 3/8, \quad q_{\min} = 107/200, \\ v_- = 15/64, \quad v_+ = 9951/40000, \quad A = 3/5, \quad d = 1/250, \\ \eta = 2/5, \quad c_{\text{mass}} = 41/100, \quad \pi_+ = 22/7, \quad \phi_- = 199/500, \quad \phi_+ = 2/5, \\ \kappa = 3, \quad \xi = 93/100, \quad c_{\text{var}} = 6, \quad \zeta = 23/100, \quad m_* = 3/4, \quad V_* = 113/1000, \\ \bar{c} = 79/200, \quad j_0 = 1/5, \quad j_1 = 5/2. \end{gathered}$$

In particular $|w| < \xi < 1$, since $(46/25)^2(v_+ + (7/20)/n_0) < \xi^2$. Also $n_0(r - d) > \xi$, so $na_0 \le t < n(1 - a_1)$.

An exact weighted crossing identity.

On an auxiliary space take independent copies of the conditional laws and write

$$T_j = j + \sum_{l \le j} U_{1l} + \sum_{l > j} U_{0l}, \qquad \mathcal{L}_j(t) = \mathbb{E}(t - T_j)_+, \qquad V_i = \sum_{l < i} U_{1l} + \sum_{l > i} U_{0l}.$$

The mesh is increasing because $T_j - T_{j-1} \geq 1 - a_0 - a_1 > 0$. Each T_j has the conditional convolution given j successes. Set

$$\chi(x) = \mathbb{E}(x - U_0)_+ - \mathbb{E}(x - 1 - U_1)_+ - \mathbb{P}(1 + U_1 \leq x), \qquad \chi_i = \mathbb{E}\chi(t - i + 1 - V_i).$$

Lemma 7.3 (Count interpolation and crossing loss)**.** *With the preceding definitions,* $\mathcal{B}_j = \mathcal{L}_j(t) - (t - j)_+ \geq 0$ *and*

$$\mathbb{P}\left(\sum X_i \leq H\right) = (1 - u)F_B(k) + uF_B(k + 1) + \sum_{j=1}^{n-1}(b_{j+1} - b_j)\mathcal{B}_j - \sum_{i=1}^{n} b_i\chi_i. \tag{39}$$

Proof. Jensen's inequality for the convex positive-part function and $\mathbb{E}T_j = j$ give $\mathcal{B}_j \geq 0$. Independence of the omitted coordinate gives $\chi_i = \mathcal{L}_{i-1}(t) - \mathcal{L}_i(t) - \mathbb{P}(T_i \leq t)$. Here $\mathcal{L}_0 = t$, $\mathcal{L}_n = 0$ and $\mathbb{P}(T_0 \leq t) = 1$: the support bounds give $T_0 \leq na_0 \leq t < T_n$. Multiplication by b_i and finite summation by parts now gives the identity, since replacing $\mathcal{L}_j$ by $(t - j)_+$ in $b_0 + b_1 t + \sum(b_{j+1} - b_j)\mathcal{L}_j$ telescopes to the stated interpolation. All CDF events are closed, including atoms. □

The functions $\mathcal{L}_j(t)$ are expectations of the convex positive-part function; their Jensen gaps $\mathcal{B}_j$ measure how noise changes those expectations. Thus the three terms in (39) are the interpolated binomial probability, an error from variation in the binomial masses, and the crossing loss. The next argument proves $\chi_i \geq 0$ using the separated conditional supports and conditional centering; this does not follow merely from $\mathcal{B}_j \geq 0$.

Five kernels with four centered replacements.

The sine bump below serves three purposes. Its translates recover a single periodic function, its Lipschitz derivative makes centered replacement errors proportional to variance, and the identity for $\sin^2$ turns independence into a product of characteristic functions. Five translates will supply the main loss; the omitted translates are bounded using a fourth moment. Let $\psi(x) = \sin^2(\pi x)$ on $[0, 1]$ and zero outside. It is continuously differentiable with derivative Lipschitz constant $2\pi^2$.

Lemma 7.4 (A nonnegative crossing kernel)**.** *For every real* x*,*

$$\chi(x) \geq \eta\psi(x) \geq 0. \tag{40}$$

Proof. For $x < 1 - a_1$, $\chi(x) = \mathbb{E}(x - U_0)_+ \geq 0$, and Jensen gives $\chi(x) \geq x$ when $x \geq 0$. On $[0, 1]$, $\psi(x) \leq (5/2)x$: use $\pi^2 < 10$ up to $1/4$, the concave tangent $1/2 + \pi(x - 1/4)$ on $[1/4, 2/5]$, and $\psi \leq 1$ thereafter. For $x \geq 1 - a_1 > a_0$, put $D_1 = 1 + U_1$ and $L_1 = \mathbb{E}(D_1 - x)_+$. Put–call parity gives $\chi(x) = \mathbb{P}(D_1 > x) - L_1 \geq 0$, since $D_1 \leq 1 + A$ and $x > A$. When $1 - a_1 \leq x \leq 1$, $L_1 \geq 1 - x$ and

$$\chi(x) \geq \frac{x - A}{1 + A - x}L_1 \geq \frac{1 - A - d}{A + d}(1 - x) \geq \eta\pi^2(1 - x)^2 \geq \eta\psi(x).$$

The penultimate inequality uses $\eta\pi_+^2 d \leq (1 - A - d)/(A + d)$. At $x = 0, 1$ both comparisons are weak, as required for atoms. □

The useful feature of the sine bump is its phase-uniform average: for any real random variable V,

$$\mathbb{E}\sin^2(\pi(u - V)) = \frac{1 - \Re\{e^{2\pi iu}\mathbb{E}e^{-2\pi iV}\}}{2} \geq \frac{1 - |\mathbb{E}e^{2\pi iV}|}{2}.$$

The characteristic function factorizes for a sum of independent noises. To use this observation, we compare five neighboring count sums with one fixed sum, paying for four centered replacements.

Lemma 7.5 (Five-count crossing loss)**.** *Under the larger-noise hypotheses,*

$$\sum_{i=k-1}^{k+3} \chi_i \ge \eta \left\{ \frac{1-e^{-\kappa(1-6/n)\lambda}}{2} - \frac{A^2(1+3/n)\lambda + 3(1+3/n)^2\lambda^2}{16} - \frac{4\pi^2\lambda}{nr} \right\}. \tag{41}$$

Proof. Fix $V = V_{k+1}$ and let $V^{(j)}$ have the law of V_{k+1+j}. For $j > 0$ it replaces j copies of U_0 by U_1. With $\psi_j(x) = \psi(u - j - x)$, regroup the two positive changes as

$$\begin{aligned} &\mathbb{E}\psi_1(V^{(1)}) + \mathbb{E}\psi_2(V^{(2)}) - \mathbb{E}\psi_1(V) - \mathbb{E}\psi_2(V) \\ &\quad = \mathbb{E}(\psi_1 + \psi_2)(V^{(1)}) - \mathbb{E}(\psi_1 + \psi_2)(V) + \mathbb{E}\psi_2(V^{(2)}) - \mathbb{E}\psi_2(V^{(1)}). \end{aligned}$$

There is an identical grouping for the negative changes. Adjacent bumps have disjoint open supports and zero derivatives at the joins, so the derivative Lipschitz constant remains $2\pi^2$. For a centered Z, $|\mathbb{E}F(a+Z) - F(a)| \le \frac{1}{2}\operatorname{Lip}(F')\mathbb{E}Z^2$. Conditioning on all unchanged summands, and applying this twice to each replacement, bounds the total cost by $4\pi^2(s_0+s_1)$. Hence

$$\sum_{i=k-1}^{k+3} \chi_i \ge \eta \left\{ \sum_{j=-2}^{2} \mathbb{E}\psi(u-j-V) - 4\pi^2(s_0+s_1) \right\}.$$

These five indices are interior, because $|i - np| < \xi + 3 < 4$. The summands within each sum are independent; neighboring count sums share the unchanged summands in the replacement argument.

The infinite translates of ψ sum to $\sin^2(\pi(u-V))$. An omitted nonzero translate requires $|V| > 2$, including endpoint collisions because the bump vanishes there. Thus the omitted part is at most $\mathbb{E}V^4/16$. With $\delta = k - np \in (-1-\xi, \xi)$, the variance of the fixed sum is

$$\sigma_V^2 = (nq - \delta - 1)s_0 + (np + \delta)s_1, \qquad (1-6/n)\lambda \le \sigma_V^2 \le (1+3/n)\lambda.$$

The lower estimate uses $c_{\mathrm{var}} r \ge 1 + \xi$; the upper uses $r \ge 1/3$. Also $s_0 + s_1 \le \lambda/(nr)$ and $\mathbb{E}V^4 \le A^2\sigma_V^2 + 3\sigma_V^4$. For either conditional noise Z, an independent copy satisfies $|Z - Z'| \le A + d < 2/3$. Concavity of sine and $\sin(2\pi/3) = \sqrt{3}/2 > 17/20$ give $\sin(\pi x) \ge (51/40)x$ on $[0, A+d]$; hence $1 - \cos(2\pi x) \ge 3x^2$ on $|x| \le A + d$. Therefore

$$1 - |\mathbb{E}e^{2\pi i Z}|^2 = \mathbb{E}\{1 - \cos(2\pi(Z - Z'))\} \ge 2\kappa \operatorname{Var} Z,$$

and $|\mathbb{E}e^{2\pi i Z}| \le e^{-\kappa \operatorname{Var} Z}$. Multiplying over the independent summands of V shows $\mathbb{E}\sin^2(\pi(u - V)) \ge (1 - e^{-\kappa\sigma_V^2})/2$. Combining the replacement cost, the omitted translates, and the characteristic-function bound proves the displayed inequality. □

The weights and the complete curvature tail.

For $k-1 \le i \le k+3$, the segment between i/n and p has variance product greater than ζ; indeed $\zeta < \min_{z=p_- - 4/n_0,\, p_+ + 4/n_0} z(1-z)$. Relative entropy is bounded by $(i-np)^2/(2n\zeta) < 8/(n\zeta)$, and the factorial remainder is greater than $-1/(12n\zeta)$. Stirling therefore gives

$$\sqrt{n}\, b_i > 2\phi_- \left(1 - \frac{8 + 1/12}{n_0\zeta}\right) > m_*.$$

Every other χ_i is nonnegative. The right-hand side of (41) increases with n. Replace n by n_0 first, then use $1-e^{-x} \ge x - x^2/2 + x^3/6 - x^4/24$ for $x \ge 0$. Writing $b = 1 + 3/n_0$ and $\kappa_0 = \kappa(1 - 6/n_0)$, we obtain

$$\sqrt{n}\sum_i b_i\chi_i \ge \mathcal{W}(\lambda),$$
$$\mathcal{W}(\lambda) := \eta m_* \left\{ \frac{\kappa_0\lambda - \kappa_0^2\lambda^2/2 + \kappa_0^3\lambda^3/6 - \kappa_0^4\lambda^4/24}{2} - \frac{A^2b\lambda + 3b^2\lambda^2}{16} - \frac{4\pi_+^2\lambda}{n_0 r} \right\}. \tag{42}$$

Lemma 7.6 (Variation of the binomial weights)**.** *Put $c = (np - q - t)_+ \le \bar{c}$. If T and M_4 bound respectively the second and fourth moments of W_j whenever $b_{j+1} > b_j$, then*

$$\sqrt{n}\sum_j (b_{j+1} - b_j)\mathcal{B}_j \le \frac{c_{\mathrm{mass}}}{nv^{3/2}} \left\{ \frac{T}{4} + c\sqrt{T} + \frac{27M_4}{256}\left(\frac{5}{3} + \frac{5c}{4}\right) \right\}. \tag{43}$$

Proof. For $a = t - j$ and $W_j = T_j - j$, centering gives

$$\mathcal{B}_j = (\mathbb{E}|a - W_j| - |a|)/2 \ge 0, \qquad \mathcal{B}_j \le \sqrt{T}/2, \qquad a_+\mathcal{B}_j \le T/4,$$

whenever $T \ge \mathbb{E}W_j^2$. The factorization $27x^4 - 256x + 256 = (3x-4)^2(3x^2+8x+16)$ gives $\mathcal{B}_j \le 27M_4/(256|a|^3)$ for $a \ne 0$, if $M_4 \ge \mathbb{E}W_j^4$. The exact adjacent-mass formula is

$$b_{j+1} - b_j = \frac{np - q - j}{(n+1)v}\,\mathbb{P}\{\mathrm{Bin}(n+1, p) = j+1\}.$$

Discard its nonpositive terms and bound the positive numerator by $(a+c)_+$. Since $c \le \bar{c} < 1$, at most one mesh point lies in each of $(-c, 0)$ and $[0, 1)$; together they cost $c\sqrt{T} + T/4$. The rest is bounded by the complete series $\sum i^{-2} < 5/3$, $\sum i^{-3} < 5/4$. Using (30) and $\sqrt{n}/(n+1)^{3/2} \le 1/n$ proves the asserted bound. □

A positive adjacent mass requires $j < np$, where its actual variance is at most $\lambda/q \le T = \lambda/q_{\min}$ and $\mathbb{E}W_j^4 \le M_4 = A^2T + 3T^2$. Moreover $L_{\mathrm{var}}(\lambda) = j_1\lambda + j_0 \ge \sqrt{T}$, since $4j_0j_1 \ge 1/q_{\min}$.

Both signs of the normal correction.

Apply (29) at $k, k+1$ and interpolate. For $G(x) = \Phi((x - np)/\sqrt{nv})$, $G''(x) \le \phi(0)(np - x)_+/(nv)^{3/2}$. The interpolation remainder is

$$\sqrt{n}\{(1-u)G(k) + uG(k+1) - G(t)\} \le \frac{\phi(0)}{2nv^{3/2}}u(1-u)(u-w)_+.$$

If $H \le 0$, changing normal variance from v to $v + s$ is favorable. Using $u(1-u) \le 1/4$ and $u^2(1-u) \le 4/27$, the total normal cost is at most $\phi_+(\xi/8 + 2/27)/(nv^{3/2})$, and $c = (-q - w)_+ \le \bar{c}$. If $H \ge 0$, $c = 0$, interpolation costs at most $\phi_+(2/27)/(nv^{3/2})$, and the variance change costs at most $\phi(0)ws/(2v^{3/2})$ by $1 - (1+x)^{-1/2} \le x/2$. The cubic correction is favorable by (31).

Define the nonnegative quadratics

$$S_c = \frac{T}{4} + cL_{\mathrm{var}}(\lambda) + \frac{27M_4}{256}\left(\frac{5}{3} + \frac{5c}{4}\right), \qquad E_- = \frac{c_{\mathrm{mass}}S_{\bar{c}} + \phi_+(\xi/8 + 2/27)}{V_*},$$
$$E_+ = \frac{c_{\mathrm{mass}}S_0 + \phi_+(2/27 + \xi\lambda/2)}{V_*}.$$

Because $V_*^2 < v_-^3$, combining (39)–(43) with these normal bounds makes the full excess strictly less than $-\mathcal{W}(\lambda) + E_\pm(\lambda)/n$, with the sign chosen according to H. A single envelope suffices for both signs, because

$$\begin{aligned} V_*(E_- - E_+) &= c_{\mathrm{mass}}\bar{c}L_{\mathrm{var}}(\lambda) + c_{\mathrm{mass}}\frac{27M_4}{256}\frac{5\bar{c}}{4} + \phi_+\xi\left(\frac{1}{8} - \frac{\lambda}{2}\right) \\ &= \frac{7889}{100000} + \frac{1751}{8000}\lambda + c_{\mathrm{mass}}\frac{27M_4}{256}\frac{5\bar{c}}{4} > 0 \qquad (\lambda \geq 0). \end{aligned}$$

Set $E = E_-$. The full excess is strictly less than $-\mathcal{W}(\lambda) + E(\lambda)/n$ for either sign of H.

Concavity reduces the interval to two endpoints.

Set $\Pi(\lambda) = \mathcal{W}(\lambda) - E(\lambda)/n_0 - \lambda/100$. The nonnegative polynomial E is a convex quadratic, whereas direct differentiation gives

$$\mathcal{W}''(\lambda) = -\eta m_* \left\{ \frac{\kappa_0^2}{4}\left[(\kappa_0\lambda - 1)^2 + 1\right] + \frac{3b^2}{8} \right\} < 0.$$

Thus Π is strictly concave. Substitution of the rational constants above gives the exact rational lower bounds

$$\Pi(1/100) > .002314, \qquad \Pi(7/20) > .061304.$$

Both are positive, proving $\Pi > 0$ on the whole closed interval. Since $n \geq n_0$ and $E \geq 0$, this proves (28) for $1/100 \leq \lambda \leq 7/20$. The endpoint calculations are rational after using $199/500 < \phi(0) < 2/5$ and $25/8 < \pi < 22/7$. Concavity supplies the bounds between the endpoints, while (36) and (43) retain all entering counts.

The two positive-noise proofs have identical support and probability hypotheses, include their shared endpoint, and cover every phase. At $\lambda = 0$, conditional centering and $\mathbb{E}U^2 = 0$ give $U = 0$ almost surely, so (29) applies directly. This completes the proof of Theorem 7.1.

8 Composition and the finite boundary

Theorem 8.1 (Exclusion of a nonnegative-excess maximizer)**.** *Let $n \geq 700$ and let (P, H) attain the fixed-target maximum at C_{E} with $a(P, H) \geq 0$. Suppose $C_{n-1} \leq C_{\mathrm{E}}$, the structural inequality* (50)*, and the complete entry Conditions D.3 and D.5 hold. Then such a pair cannot exist.*

Proof. Theorem 5.1 gives $nD \leq 32.20665378$ and $|H| \leq 1.83605680$ for this same compactly supported law. Propositions 6.1 and 6.2 give the central gap and the following bounds. In (6a), identify the local theorem's B with $B_0 = \mathbf{1}_{\{X>0\}}$. The variable $(B + U - p)/\sqrt{pq + s}$ is exactly the original X, so the threshold remains the original raw H. The hypotheses match as follows.

Bound just obtained	Hypothesis of Theorem 7.1
$3/8 < p < 93/200$	$p \in [3/8, 93/200]$
$a_0, a_1 \geq 0$, $a_0 + a_1 < 1/250$	Nonnegative inward endpoints, sum at most $1/250$
$-3/5 < U_0 \leq a_0$, $-a_1 \leq U_1 < 3/5$	The stated closed conditional-support bounds
$0 \leq \lambda < 7/20$	Positive-noise interval or separate zero-noise case
$\lvert H\rvert \leq 1.83605680 < 46/25$	The full threshold interval
Independent copies of X	The iid sum in the local comparison

Positive noise gives $a(P, H) < -\lambda/100$; zero noise is Bernoulli and gives strictly negative excess by Schulz. Both contradict $a \geq 0$. □

Theorem 8.2 (Conditional sharp inequality)**.** *Assume a genuine finite base $C_j \leq C_{\mathrm{E}}$ for $1 \leq j \leq 699$, the external structural and strict finite-binomial theorems used above, and the complete entry predicates at every least failure at $n \geq 700$. Then*

$$C_n < C_{\mathrm{E}} \ (n \geq 700), \qquad C_n \leq C_{\mathrm{E}} \ (n \geq 1), \qquad \sup_{n \geq 1} C_n = C_{\mathrm{E}}.$$

Proof. If $C_n < C_{\mathrm{E}}$ fails above the base, choose the least failing index. Proposition 2.1 and the reflection argument in Section 2.2 supply an attained positive closed-event pair with $a \geq 0$. The finite base and minimality give the predecessor $C_{n-1} \leq C_{\mathrm{E}}$. The preceding theorem excludes the pair. The independent Bernoulli limit (2) gives equality for the global supremum. □

If only a base through 41 were known, the same argument would locate a first failure in $42, \ldots, 699$. The coarse theorem would still localize that maximizer, but the support and count comparisons at $n \geq 700$ would not apply to it. The reflection inequality, including its endpoint-atom convention, is proved in Section 2.2.

Role of finite verification.

The computational input consists of Conditions D.3 and D.5, used through Theorem 5.1. The scalar comparisons in Sections 6–7 are separate analytical inequalities with short exact checks. Appendix E gives the corresponding reproduction commands. Neither computation supplies the finite base.

9 Discussion

The fixed target determines both the third-moment penalty in the extremal problem and the coefficient in the conservation identity. The gap inequality explains their common factor $3 + \sqrt{10}$. A bound on the remainder then restricts the defect of a maximizer, and count conditioning turns that geometric restriction into a strict comparison with a binomial law.

The number 700 is a common sufficient floor for two parts of the proof. The coarse bound satisfies $32.20665378/700 < 0.047$, as required by the geometric estimates. The local comparison also uses $n \geq 700$ in its binomial-mass and error bounds. Reducing the threshold therefore requires checking both parts: a sharper localization constant alone does not establish the comparison at smaller sample sizes. A different certified localization calculation or a comparison valid on a larger law region could reduce the remaining finite range, but either would need its own proof and verification.

The small-noise estimate keeps its errors proportional to λ. The five-count estimate has an additive error of order $1/n$, so it is used on $1/100 \leq \lambda \leq 7/20$. At zero noise, the law is Bernoulli and the strict finite-binomial theorem applies. These three cases give the exclusion at $n \geq 700$ under the hypotheses stated above. The base cases through 699 are not treated here; they remain the additional requirement for the global sharp conclusion.

AI assistance.

AI systems were used extensively for mathematical exploration, exact checking programs, source and bibliography comparisons, and drafting and revision. Separate AI review roles examined component arguments; their agreement is not a substitute for mathematical proof or external human review. Failed arguments remain diagnostics, not premises. The authors are responsible for the content and presentation.

Sources.

The immutable revision snapshot contains the revised proof, scalar checkers and dated execution records in `manuscript/compact-v16/`, including `publication/opus-20261030/`. This release adds its source citation and metadata; the mathematical arguments and checkers are unchanged. The retained entry certificate is also available at the original immutable baseline. Appendix E identifies the certificate, its exact arithmetic, and the commands for checking this revision. A document build checks presentation, not theorem validity.

A A uniform Fourier enclosure with explicit input hypotheses

We prove the compensated Fourier formula and the complete low- and high-frequency estimates used in Section 5. Every assertion concerns an actual marginal with the displayed moment and event bounds. Appendix D establishes these inputs for the maximizers to which the bound is applied, under its stated entry conditions.

For each integer $n \geq N \geq 42$, let $X = X_n$ be a real random variable with $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, and $\mathbb{E}|X|^3 < \infty$. Neither the law nor the real raw threshold $H = H_n$ must be fixed as n varies. Write

$$f(t) = \mathbb{E}e^{itX}, \quad \gamma = \mathbb{E}X^3, \quad e_n(t) = e^{-nt^2/2}(1 - in\gamma t^3/6), \quad V_n = f^n - e_n.$$

The two complete integrals are

$$I_0(H) = \int_0^{1/2} \frac{|\Im(e^{-itH}V_n(t))|}{t^3}\, dt, \qquad I_\infty = \int_{1/2}^{\infty} \frac{|V_n(t)|}{t^3}\, dt.$$

A.1 The inversion edge, including atoms and moving thresholds

The sum law and its signed Gaussian approximation have the same mass and first moment. Subtracting these two terms before integration makes the Fourier expression absolutely integrable. The following elementary identity also explains why no continuity assumption at the threshold is needed.

Lemma A.1. *If a finite signed measure μ has mass and first moment zero and $\int x^2\, d|\mu|(x) < \infty$, then, for every $H \in \mathbb{R}$,*

$$\int (x - H)|x - H|\, d\mu(x) = -\frac{4}{\pi} \int_0^{\infty} \frac{\Im(e^{-itH}\widehat{\mu}(t))}{t^3}\, dt.$$

The integral on the right is absolutely convergent.

Proof. For real y, substitution and integration by parts give

$$\int_0^{\infty} \frac{ty - \sin(ty)}{t^3}\, dt = \frac{\pi}{4} y|y|.$$

The boundary terms vanish: near zero the numerator is $O(t^3)$, and at infinity it is $O(t)$. The integrand has the sign of y, since $v - \sin v \geq 0$ for $v \geq 0$. To calculate the constant, nonnegative Tonelli and elementary Laplace integration give

$$\int_0^{\infty} \frac{1 - \cos t}{t^2}\, dt = \int_0^{\infty} s\left(\frac{1}{s} - \frac{s}{s^2 + 1}\right) ds = \frac{\pi}{2}.$$

The absolute double integral of the compensated expression against $d|\mu|(x)$ equals

$$\frac{\pi}{4} \int (x - H)^2 d|\mu|(x) < \infty.$$

Absolute Fubini is therefore valid. The linear compensation disappears because $\int (x - H) d\mu = 0$, proving both assertions. No continuity of a CDF at H is involved. □

Apply this to the sum law minus the scale-$\sqrt{n}$ pushforward of the signed density

$$\phi(x)\{1+\gamma(x^3-3x)/(6\sqrt{n})\}.$$

This signed density has mass one, first moment zero and finite absolute second moment. Its transform after scaling is e_n. Gaussian integration by parts, using $q_z(x)=(x-z)|x-z|$ and $q_z'''=4\delta_z$, yields

$$\varepsilon_n := \mathbb{E}q_{H/\sqrt{n}}(S_n/\sqrt{n}) - \mathbb{E}q_{H/\sqrt{n}}(Z) - \frac{2\gamma\phi(H/\sqrt{n})}{3\sqrt{n}} = -\frac{4}{\pi n}\int_0^\infty \frac{\Im(e^{-itH}V_n(t))}{t^3}\,dt.$$

Consequently the error $E_n=-n^{3/2}\varepsilon_n/2$ satisfies

$$|E_n| \le \frac{2\sqrt{n}}{\pi}\{I_0(H)+I_\infty\}. \tag{44}$$

The identity and bound hold for each n and H separately.

A.2 Low frequencies: Taylor cancellation and moment bounds

Near zero, the Gaussian approximation cancels the first terms of the characteristic function. The fourth- and fifth-order coefficients must still be bounded, and the sixth moment controls the remaining Taylor error. These higher moments are explicit hypotheses of the estimate; in its application they become available only after support has been bounded.

Assume common bounds

$$\mu_4 := \mathbb{E}X^4 \le M \le 111/10, \quad |\gamma| \le g \le 37/20, \quad \mu_6 \le M_6,$$

and put

$$a=(3-\mu_4)/24, \quad z=(10\gamma-\mathbb{E}X^5)/120, \quad |a|\le A, \quad |z|\le Z.$$

Here A, Z, M_6 are nonnegative common ceilings, not quantities inferred from bounded third moments. Define

$$E = M_6/720 + g^2/72 + (1/2+g^2/144)^3/5, \qquad B = A + Z/2 + E/4,$$

$$a_0 = \frac{N-1}{2N}, \quad b_0 = \frac{45-M}{96}(1-2/N) > 0, \quad \lambda_N = \sqrt{\pi/2}\left(\frac{N}{N-1}\right)^{3/2}.$$

Proposition A.2. *For all the laws and thresholds above,*

$$I_0(H) \le \frac{L_0 + L_H|H|}{\sqrt{n}} + \frac{L_2}{n},$$

where, for any $c_0 \ge \lambda_N$ and $c_1 \ge \lambda_N/2$,

$$\begin{aligned} L_0 &= c_0 Z + c_1 A g + 251 g^3/17280, \\ L_H &= c_0 A + 251 g^2/4800, \\ L_2 &= E/(2a_0^2) + B^2/(2b_0^3). \end{aligned}$$

Proof. Let $h(t)=\exp(-t^2/2 - i\gamma t^3/6)$. Taylor's integral remainder for e^{itX}, followed by expectation, gives its expansion through degree five with error at most $\mu_6 t^6/720$. For $w=-t^2/2-i\gamma t^3/6$ and $0\le t\le 1/2$,

$$|w| \le t^2(1/2+g^2/144) < 1/6, \quad |e^w - 1 - w - w^2/2| \le |w|^3 e^{|w|}/6 \le |w|^3/5.$$

The first bound follows from $\sqrt{1/4+y} \le 1/2+y$; for the last use $e^{1/6} \le (1-1/6)^{-1} = 6/5$. Comparing the two polynomials gives

$$f - h = -at^4 - izt^5 + r_6(t), \qquad |r_6(t)| \le Et^6, \qquad |f-h| \le Bt^4. \tag{45}$$

The degree-six term in $w^2/2$ costs $g^2t^6/72$; it has not been discarded.

Four integrations of $1-\cos u \ge 0$ show $\cos u \le 1 - u^2/2 + u^4/24$ for every real u. Independence gives $\mathbb{E}(X-X')^4 = 2\mu_4 + 6$, and hence

$$|f(t)|^2 \le 1 - t^2 + (M+3)t^4/12 \le 1 - (45-M)t^2/48.$$

Thus $|f|, |h| \le e^{-(45-M)t^2/96}$ throughout the low interval. Every point on their complex joining segment obeys the same bound. Taylor's integral formula for the polynomial $v \mapsto v^n$ proves

$$f^n - h^n = n(f-h)h^{n-1} + R_2, \quad |R_2| \le \tfrac{1}{2}n(n-1)B^2t^8 e^{-(n-2)(45-M)t^2/96}.$$

The leading remainder in (45) and this power remainder cost at most L_2/n after division by t^3 and integration: use $n-1 \ge 2a_0 n$, $(n-2)(45-M)/96 \ge b_0 n$, $n(n-1) \le n^2$, and $\int_0^\infty t^3 e^{-At^2}dt = 1/(2A^2)$, $\int_0^\infty t^5 e^{-At^2}dt = A^{-3}$.

For the remaining leading terms let $b = n-1$ and $v = b\gamma t^3/6$. Their imaginary part after rotation by Ht is bounded by

$$ne^{-bt^2/2}\{(Z + A|H|)t^5 + bAgt^7/6\}.$$

The Gaussian moments of orders two and four, after dividing by t^3, give $n\sqrt{\pi/2}b^{-3/2}(Z + A|H| + Ag/2)$. This is at most $\lambda_N(Z + A|H| + Ag/2)/\sqrt{n}$. Finally put $u = n\gamma t^3/6$. Direct sine and cosine remainders give

$$|\Im[e^{-itH}(h^n - e_n)]| \le e^{-nt^2/2}(|u|^3/6 + |H|tu^2/2).$$

The Gaussian moments of orders six and four respectively give $\sqrt{\pi/2}g^3/(432\sqrt{n})$ times 5, and $\sqrt{\pi/2}g^2|H|/(24\sqrt{n})$. Using $\sqrt{\pi/2} < 251/200$ gives precisely the two displayed rational coefficients. Every extended integrand was nonnegative. The only index comparisons were the displayed rational inequalities; in particular the law and H can vary arbitrarily with n. □

In the selected enclosure one has actual support $|X| \le R$ with $1 \le R \le 6$ and $\mathbb{E}|X|^5 \le E_5$. Then $M_6 = RE_5$ and $A = \max(2, M-3)/24$ are valid. Its sharper Z is also licensed: for $Q = X^2 - \gamma X - 1$ direct multiplication gives

$$\mathbb{E}Q^2 = \mu_4 - 1 - \gamma^2, \quad \mathbb{E}X^5 = \gamma^3 + 2\gamma + \mathbb{E}[Q^2(X + 2\gamma)].$$

Reflect this auxiliary calculation if necessary so that $t = |\gamma| \ge 0$. For any $r \ge \max(R, 2g)$ the two signs of $\mathbb{E}X^5 - 10\gamma$ are at most

$$P_r(t) = r(M-1) + (10-2M)t - rt^2 + t^3, \quad Q_r(t) = r(M-1) + (2M-10)t - rt^2 - t^3.$$

The coefficients of μ_4 before replacing it by M are $r - 2t, r + 2t \ge 0$. If $M \le 5$, $P_r \ge Q_r$ and $P_r \le r(M-1) + (10-2M)t - (r-g)t^2$, whose maximum on $[0, g]$ is at the clipped vertex used in the code. If $M > 5$, $P_r \le r(M-1)$ and $Q_r \le r(M-1) + (2M-10)t - rt^2$, again with the stated clipped vertex. For $r = 6$, $P_6'' = -12 + 6t < 0$ on $[0, g]$. If $M \le 5$ and $P_6'(g) \ge 0$, its maximum is exactly $P_6(g)$; otherwise the preceding quadratic gives $6(M-1) + (10-2M)^2/[4(6-g)]$. For $M > 5$ the unrestrained vertex bound is $6(M-1) + (2M-10)^2/24$. Taking the minimum of these proved ceilings and $E_5 + 10g$, then dividing by 120, gives a valid coefficient ceiling Z. The coefficient ceiling therefore uses the known support radius R.

A.3 High frequencies: six contributions and uniform index bounds

A nearly lattice law can retain characteristic-function peaks away from zero. The high-frequency argument therefore uses an event involving the actual difference $X - X'$ and partitions the oscillations around their zeros. The six contributions below cover intermediate frequencies, nonzero-cell cores, nonzero-cell wings, the small-angle part of the zero cell, the rest of the zero cell, and the signed Gaussian comparator. Each optional improvement has its own stated positivity and monotonicity conditions.

For independent copies of the same marginal suppose an actual event $\mathcal{E}$ satisfies

$$\mathbb{P}(\mathcal{E}) \geq \alpha > 0, \quad |X - X'| \geq d > 0 \text{ on } \mathcal{E}, \quad \mathbb{E}[(X - X')^2 \mid \mathcal{E}] \leq M_E.$$

Fix $b = 9/10$, $0 < r \leq 1/4$, and

$$h = 333/106 - (22/7)^3 r^2/6 > 0, \quad \kappa = \alpha(23/48)^2 d^2, \quad L(x) = \frac{x+2}{x^2+4x+2},$$

$$S_3 = \sum_{j=1}^{20} j^{-3} + 1/800, \quad S_5 = \sum_{j=1}^{20} (j-r)^{-5} + \frac{1}{4(20-r)^4}.$$

Let $c(u) = u/2 - (M+3)u^2/24$ and set

$$c_* = \min\{c(1/4), c(81/100)\}, \quad c_q = 1/8 - (M+3)/384, \quad s = [1 - (M+3)53/600]/2.$$

Proposition A.3. *Suppose $c_* > 0$. For every $n \geq N$, $\sqrt{n} I_\infty$ is at most the sum of the following six terms:*

$$\begin{aligned}
T_1(n) &= 2\sqrt{n} e^{-nc_*}, \\
T_2(n) &= \frac{M_E}{4\pi^{3/2} h \sqrt{\alpha}} \{S_3 + 3S_5/(n\alpha h^2)\}, \\
T_3(n) &= \frac{M_E \sqrt{n}}{2\pi^2} e^{-n\alpha h^2 r^2}, \\
T_4(n) &= \frac{e^{-n\kappa b^2}}{2b^4 \kappa \sqrt{n}}, \\
T_5(n) &= \frac{\sqrt{n}}{2b^2} e^{-n\alpha(23/48)^2}, \\
T_6(n) &= \sqrt{n} e^{-n/8} \{2L(n/8) + \sqrt{M-1}/3\}.
\end{aligned}$$

If $s > 0$, T_1 may be replaced by its minimum with $2\sqrt{n} e^{-nc_q} L(ns/4)$. The fourth term may be replaced by its minimum with $\sqrt{n} e^{-n\kappa b^2} L(n\kappa b^2)/(2b^2)$. The common-floor sum at N is valid if

$$2Nc' > 1 \quad \text{for } c' \in \{c_*, \alpha h^2 r^2, \alpha(23/48)^2, 1/8\}.$$

For the optional first term also require $2Nc_q > 1$; for the optional fourth term require $N\kappa b^2 \geq 1/4$. If an optional gate fails, the simpler term remains available.

Proof. The triangle inequality separates $|f^n|$ and the entire comparator. On $[1/2, b]$, symmetrization gives $|f|^n \leq e^{-nc(t^2)}$. The function c is concave, so $c(t^2) \geq c_*$ and the integral of t^{-3} is less than two. This gives T_1. Also

$$c(u) - c_q - s(u - 1/4) = (M+3)(u - 1/4)(81/100 - u)/24 \geq 0.$$

When $s > 0$ this chord may be integrated to infinity.

For completeness, the tail calculation used here is, for $x = nab^2 > 0$,

$$\sqrt{n}\int_b^\infty e^{-nat^2}t^{-3}dt \le \frac{\sqrt{n}e^{-x}}{2b^2}L(x). \tag{46}$$

To see where the rational function L comes from, let U have density e^{-u} on $u \ge 0$. Substitution followed by positive Fubini gives

$$\sqrt{n}\int_b^\infty e^{-nat^2}t^{-3}dt = \frac{\sqrt{n}e^{-x}}{2b^2}\mathbb{E}\frac{U}{x+U}.$$

For example, after putting $v = na(t^2 - b^2)$ the remaining factor is $\int_0^\infty xe^{-v}(x+v)^{-2}dv$; integration by parts makes it $\mathbb{E}[U/(x+U)]$. For $x > 0$ put

$$J_x = \mathbb{E}\frac{1}{x+U}, \qquad v_0 = \begin{pmatrix}1\\1\end{pmatrix}, \qquad G = \mathbb{E}\left[(x+U)\begin{pmatrix}1 & U\\ U & U^2\end{pmatrix}\right] = \begin{pmatrix}x+1 & x+2\\ x+2 & 2x+6\end{pmatrix}.$$

The entries use $\mathbb{E}U = 1$, $\mathbb{E}U^2 = 2$, and $\mathbb{E}U^3 = 6$. For $q(U) = c_1 + c_2U$, weighted Cauchy–Schwarz in its squared-residual form gives

$$0 \le \mathbb{E}\frac{[1-(x+U)q(U)]^2}{x+U} = J_x - 2v_0^Tc + c^TGc.$$

The matrix G is positive definite: its leading entry is positive and $\det G = x^2 + 4x + 2 > 0$. Choosing $c = G^{-1}v_0$ yields

$$J_x \ge v_0^TG^{-1}v_0 = \frac{x+3}{x^2+4x+2}, \qquad \mathbb{E}\frac{U}{x+U} = 1 - xJ_x \le \frac{x+2}{x^2+4x+2} = L(x).$$

This proves (46) for every $a, b, n > 0$ with the same function L. Applying it at endpoint $1/2$ and rate s proves the optional first term.

For $t \ge b$, symmetrization, the actual-event probability bound, and convex Jensen give

$$|f(t)|^n \le e^{-n\mathbb{E}\sin^2(t(X-X')/2)} \le \mathbb{E}[e^{-n\alpha\sin^2(tD/2)} \mid \mathcal{E}], \quad D = |X - X'|.$$

Positive Tonelli allows a separate partition for every $D > 0$. With $\ell = 2\pi/D$, use the zero cell $[0, \ell/2]$ and nonzero cells $\ell[j - 1/2, j + 1/2]$, $j \ge 1$. In a nonzero cell write $t = \ell(j + v)$. On its core $|v| \le r$, $|\sin \pi v| \ge h|v|$; on its wings $r \le |v| \le 1/2$ it is at least hr. These follow from $\sin x \ge x - x^3/6$ and monotonicity of sin on $[0, \pi/2]$, with directed bounds on π. For $|v| \le r$ Taylor's integral remainder gives

$$\{(j-v)^{-3} + (j+v)^{-3}\}/2 \le j^{-3} + 6(j-r)^{-5}v^2.$$

Extend the nonnegative Gaussian majorant over the whole real line. Its zeroth and second moments, the Jacobian $t^{-3}dt = D^2(4\pi^2)^{-1}(j+v)^{-3}dv$, and $\mathbb{E}[D^2 \mid \mathcal{E}] \le M_E$ give T_2. The two series tails are bounded by their decreasing integrals after 20. The total wing weight is at most $\int_{\ell/2}^\infty t^{-3}dt = D^2/(2\pi^2)$, giving T_3. Extending these sets below b only increases their nonnegative bounds.

In the zero cell put $u = tD/2 \le \pi/2$. Where $u \le 1/2$, $\sin u \ge 23u/24$, and $D \ge d$ yields rate κ. Extending the set to $[b, \infty)$ gives T_4 from $t^{-3} \le t/b^4$; (46) gives its optional improvement. Where $u \ge 1/2$, monotonicity gives $\sin u \ge 23/48$ and the total weight is at most $1/(2b^2)$, proving T_5. Endpoints have zero Lebesgue measure even if the conditional distance distribution has atoms.

Finally $|e_n(t)| \le e^{-nt^2/2}(1 + n|\gamma|t^3/6)$. The first part on the entire $[1/2, \infty)$ uses (46); the second uses $\int_{1/2}^\infty e^{-nt^2/2}dt \le 2e^{-n/8}/n$. Pearson's identity $\mu_4 - 1 - \gamma^2 = \mathbb{E}(X^2 - \gamma X - 1)^2 \ge 0$ gives T_6. No comparator tail is dropped.

For floor evaluation, $\sqrt{n}e^{-nc'}$ decreases for $2nc' > 1$, whereas n^{-1} and $n^{-1/2}e^{-nc'}$ decrease for every $c' > 0$. Also $L'(x) = -(x^2 + 4x + 6)/(x^2 + 4x + 2)^2 < 0$. Thus the optional first term decreases under its own gate, and the comparator decreases for $n \geq 5$. The sign of the derivative of $\sqrt{x}e^{-x}L(x)$ is the sign of

$$4 - 10x - 22x^2 - 13x^3 - 2x^4,$$

which is negative at $1/4$ and strictly decreasing on $(0, \infty)$. This proves the optional zero-cell gate. Every comparison is of a common scalar majorant, not of characteristic functions for different laws. □

A.4 The precise closed Fourier conclusion

Under the preceding moment, actual-event and scalar gates, let K_H be any complete HIGH sum evaluated at N, including its permitted optional improvements. Then

$$|E_n| \leq \frac{2}{\pi}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\} \leq \frac{16}{25}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\}. \tag{47}$$

The second inequality uses $\pi > 25/8$ and nonnegative coefficients. This is the constant Fourier budget and raw-threshold slope used in Section 5. Complete HIGH bounds for several valid windows may be minimized. The LOW coefficients must remain one complete valid triple.

The rational enclosures $\pi > 333/106$, $\sqrt{\pi} > 4431/2500$, $\pi < 22/7$ and $\sqrt{\pi/2} < 251/200$ are proved together in Appendix B. The arithmetic routines and their rounding directions are described in Appendix E. The floor comparisons concern the analytical majorants under their stated positivity and monotonicity conditions; the rounded routines need not themselves be monotone. The resulting bounds hold for each law and threshold satisfying the hypotheses, even when they vary with n.

B Elementary Gaussian constants

Lemma B.1 (Elementary Gaussian constants)**.** *For the standard normal density,*

$$\|\phi''\|_\infty = \phi(0) < 2/5, \qquad \|\phi'''\|_\infty \leq 2/\pi, \qquad 1/(4\pi) < 2/25.$$

Moreover $819/2000 < C_{\mathrm{E}} < 20487/50000 < 41/100$ *and* $\phi(0) > 0.39886 > 0.3988 = 997/2500$.

We give the exact comparisons used in Lemma B.1.

Proof. Since $\phi''(x) = (x^2 - 1)\phi(x)$, the norm on $x^2 \leq 1$ is at most $\phi(0)$. On $x^2 \geq 1$, the maximum of $(x^2 - 1)e^{-x^2/2}$ is $2e^{-3/2} < 1$, attained when $x^2 = 3$. Thus the norm is exactly $\phi(0)$.

For completeness, the Gaussian transform can be obtained without an inversion theorem for the distribution being studied. Differentiating $J(t) = \int_{\mathbb{R}} e^{-x^2/2}e^{itx}\,dx$ under the integral and integrating by parts give $J'(t) = -tJ(t)$. The Gaussian integral, evaluated by squaring it and using polar coordinates, gives $J(0) = \sqrt{2\pi}$. Hence $J(t) = \sqrt{2\pi}e^{-t^2/2}$; applying the same formula with the variables interchanged gives

$$\phi(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-t^2/2}e^{-itx}\,dt.$$

Three differentiations are justified by the integrability of $|t|^3e^{-t^2/2}$. Consequently

$$\|\phi'''\|_\infty \leq \frac{1}{2\pi}\int_{\mathbb{R}} |t|^3 e^{-t^2/2}\,dt = \frac{2}{\pi}.$$

The last integral equals four after the substitution $w = t^2/2$ on the positive half-line.

Here are elementary rational comparisons for all numerical constants used in this section. Put $a = \arctan(1/5)$ and $b = \arctan(1/239)$. Tangent addition gives $\tan(2a) = 5/12$, $\tan(4a) = 120/119$ and $\tan(4a - b) = 1$. Also

$$0 < 4(1/5 - 1/(3 \cdot 5^3)) - 1/239 < 4a - b < 4/5 < 1 < \pi/2.$$

The arctangent inequalities follow by integrating $1 - t^2 < 1/(1 + t^2) < 1$, while $\pi/2 = 2\arctan(1) > 1$ follows by integrating $1/(1 + t^2) > 1/2$ on $[0, 1)$. It follows that $\pi = 16a - 4b$ and

$$\pi > 16(1/5 - 1/(3 \cdot 5^3)) - 4/239 = 281476/89625 > 25/8.$$

This proves $\phi(0) < 2/5$ and $1/(4\pi) < 2/25$.

To bound C_{E} more closely, define

$$s_k(x) = \sum_{j=0}^{k-1} \frac{(-1)^j x^{2j+1}}{2j+1}, \quad \pi_- = 16s_6(1/5) - 4s_1(1/239), \quad \pi_+ = 16s_5(1/5) - 4s_2(1/239).$$

The remainder in the finite geometric expansion of $1/(1 + t^2)$ has its stated alternating sign, so integration gives $\pi_- < \pi < \pi_+$. Let $r_- = 3162277660/10^9$ and $r_+ = 3162277661/10^9$. Squaring shows $r_-^2 < 10 < r_+^2$. The following comparisons are exact rational ones:

$$\left(\frac{819}{2000}\right)^2 < \frac{19 + 6r_-}{72\pi_+} < C_{\mathrm{E}}^2 = \frac{19 + 6\sqrt{10}}{72\pi} < \frac{19 + 6r_+}{72\pi_-} < \left(\frac{20487}{50000}\right)^2.$$

For example, each outer comparison follows by multiplying by its positive rational denominator; the displayed finite sums specify those denominators completely. All quantities are positive, so taking square roots proves the claimed bounds. In particular, $0.4095 < C_{\mathrm{E}} < 0.40974 < 0.41$. The same exact endpoints also satisfy

$$\pi_- > \frac{333}{106} > \left(\frac{4431}{2500}\right)^2, \qquad \pi_+ < \frac{22}{7} < 2\left(\frac{251}{200}\right)^2.$$

Thus $\sqrt{\pi} > 4431/2500$ and $\sqrt{\pi/2} < 251/200$, as used in the Fourier bounds. Since $(0.39886)^2 < 7/44$, we also obtain $\phi(0) > 0.39886 > 0.3988$. □

Lemma B.2 (The narrow entry enclosure)**.** *The target constant satisfies*

$$c_- := \frac{4097321837023}{10^{13}} < C_{\mathrm{E}} < \frac{4097321837025}{10^{13}} =: c_+.$$

Proof. Use the same finite sums s_k and Machin identity proved above, now with

$$\pi_L = 16s_{10}(1/5) - 4s_5(1/239), \qquad \pi_U = 16s_9(1/5) - 4s_4(1/239).$$

An even number of terms bounds each arctangent from below and an odd number bounds it from above. The negative coefficient of the second arctangent reverses its choice of endpoint, so $0 < \pi_L < \pi < \pi_U$. Put

$$r_L = \frac{3162277660168379}{10^{15}}, \qquad r_U = \frac{3162277660168380}{10^{15}}.$$

Integer squaring gives $r_L^2 < 10 < r_U^2$. Direct rational comparisons give

$$c_-^2 < \frac{19 + 6r_L}{72\pi_U} < C_{\mathrm{E}}^2 < \frac{19 + 6r_U}{72\pi_L} < c_+^2.$$

The finite sums specify every rational in this display; each outer inequality can be checked by cross-multiplying positive integers. All terms are positive, so taking square roots gives the assertion. This directed enclosure, rather than the coarser bounds in the preceding lemma, supplies the precision used by the entry calculation. □

C The identity without contact

The main proof uses the contact identity. This stronger extension applies to every compactly supported standardized law and also provides an identity for checking numerical implementations on nonextremal laws. Retain the definitions of $a, Q, h, \Delta, J, J_1, J_2, L$, and $\mathcal{R}_c$ from the main text, for arbitrary $H \in \mathbb{R}$ and $c > 0$. Introduce the contact residuals

$$\xi = \mathbb{E}[X\Delta(X)], \qquad R_{\mathrm{supp}} = \int_K \tau(t)\Delta(t)\,dt + \sum_{(u,v)} \tau_{u,v}(v-u)\Delta(v). \tag{48}$$

The first integral is with respect to Lebesgue measure on the closed support K, not to P. Since Δ is bounded on the convex hull and $\sum \tau_{u,v}(v-u) \le 1$, the series is absolutely convergent.

Theorem C.1 (Conservation with contact residuals)**.** *For every such law, threshold, and positive coefficient,*

$$(n+H^2)a + nc\rho - \frac{n\phi_z}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c + n^{3/2}(R_{\mathrm{supp}} - H\xi). \tag{49}$$

At a fixed-target maximizing pair, both residuals vanish.

Proof. The gapwise identity in the proof of Theorem 4.1 holds without contact. Keeping its endpoint terms and the integral over K adds exactly R_{supp} to the right side of (19). The zero-bias evaluation (20) is unchanged. In the first-moment calculation, $\mathbb{E}[Xh(X)] = \mathbb{E}[XQ(X)] - \xi$, so (21) becomes

$$M + \phi_z = -\frac{\gamma H \phi_z}{2n} + \frac{cK_1}{n} - \sqrt{n}\xi.$$

The same pointwise identity for q_z therefore gives

$$\sqrt{n}\mathcal{A} = c\rho + a - \gamma\phi_z/6 + P_0/n + \sqrt{n}\varepsilon/2 + H\sqrt{n}\xi,$$

with exactly the P_0 defined in the main proof. Equating the two expressions and multiplying by n gives (49). At contact, $\Delta = 0$ on all of K, so every term in both residuals vanishes, including the Lebesgue integral. □

Why the support integral is needed.

Let X be uniform on $[-\sqrt{3}, \sqrt{3}]$ and take $H = 0$. There are no support gaps, $F = T = 1/2$, $\beta = \gamma = 0$, and $K_q = 3\sqrt{3}/40$. Thus

$$R_{\mathrm{supp}} = \int_K \tau\Delta = cK_q/n^{3/2} > 0.$$

Dropping the support integral would give a false identity even though all gap sums are empty. This example tests the extension without contact; it does not contradict the contact theorem. The leakage integral retains $[u, v)$ throughout: a right-endpoint atom belongs to $h(v)$, and a left-endpoint atom has weight zero.

D Details of the coarse localization

This appendix proves the analytical estimates and specifies the finite conditions used by Theorem 5.1. The calculation has two coarse quadratic passes followed by four signed-moment passes. Each improvement is used only after its hypotheses have been established. The cover starts with 24 closed intervals for ρ. A signed pass splits each surviving interval into

8 intervals for β, 8 for γ and 2 for m. Four such passes process 11264 nominal boxes; the number is smaller than $4 \cdot 24 \cdot 128$ because excluded moment cells need no later pass. Closed shared endpoints belong to both neighboring intervals. Appendix E describes what the certificate stores and how this cover is reconstructed. Let P be a centered, unit-variance law, S_j a sum of j independent copies, and set

$$\rho = \mathbb{E}|X|^3, \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3, \quad c = C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}.$$

The threshold H always has the units of the raw sum. A fixed-target maximizer maximizes

$$a(P,H) = \sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n})\} - c\rho.$$

We use Proposition 2.1 and its closed-support contact formula (4). In particular its closed-support contact equality, not ratio optimality, is the variational premise here. We consider a maximizer with $a \ge 0$. The actual normalized score is $c + a/\rho$; no upper bound on it is assumed.

D.1 The first support bound

Theorem D.1 (Initial moment and support entry)**.** *Assume $n \ge 42$, $C_{n-1} \le c$, and the universal structural inequality*

$$\sqrt{n}\,\Delta_n(P) \le \frac{3031}{10000}\left(\rho + \frac{323}{500}\right) \tag{50}$$

for standardized laws with finite third moment. Then a fixed-target maximizer with $a \ge 0$ satisfies

$$1 \le \rho < \frac{7361}{4000} < \frac{1841}{1000}, \qquad |\beta| < \frac{321}{500}, \qquad \operatorname{supp} P \subset (-96/25, 96/25). \tag{51}$$

The theorem is uniform in all real H and in both signs of a support point. It requires neither C_{n-2} nor an excess bound of order $1/n$.

The structural inequality (50) is the external iid theorem of Shevtsova [14]. Only this affine bound is needed for the displayed moment radius; it is not a consequence of the finite arithmetic in this section.

Proof. Since $c\rho \le \sqrt{n}\Delta_n(P)$ and $c > 819/2000$, elementary division gives

$$\rho < \frac{(3031/10000)(323/500)}{819/2000 - 3031/10000} = \frac{7361}{4000}.$$

The other affine bound $0.3322(\rho + 0.429)$ alone would give the larger number $712569/386500$; it must not be used to claim $7361/4000$. Lemma 2.4 gives

$$\rho \ge \frac{1+\beta^2}{\sqrt{1-\beta^2}}.$$

This function increases with $|\beta|$. For $r = 1841/1000$, $b = 321/500$,

$$(1+b^2)^2 - r^2(1-b^2) = \frac{93288849}{50000000000} > 0,$$

which proves the second assertion without assuming bounded support.

Write $v = n-1$, $s = \sqrt{n/v}$, $F_v(x) = \Phi(x/\sqrt{v})$ and $\eta_n = n(s-1) - 1/2$. Closed-support contact, the upper predecessor bound $F_{n-1,P}(H-t) \le F_v(H-t) + c\rho/\sqrt{v}$, and $\sqrt{n}(F_{n,P}(H) - \Phi(H/\sqrt{n})) = c\rho + a$ give

$$0 \le H_n(H,t) + c\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho\eta_n\} - na \quad (t \in \operatorname{supp} P), \tag{52}$$

where exactly

$$\frac{H_n(H,t)}{n^{3/2}} = F_v(H-t) - F_{v+1}(H) + tF'_{v+1}(H) - \frac{t^2-1}{2}F''_{v+1}(H).$$

The extra $-na$ can be discarded in this upper bound. This is the precise reason the predecessor assumption is $C_{n-1} \le c$, not a comparison with the actual score. No continuity assumption on the distribution function at $H-t$ is inserted.

Rationalization gives

$$n\eta_n = \frac{s^2(2s+1)}{2(s+1)^2}, \qquad s < 81/80 =: S, \qquad n\eta_n < \frac{793881}{2073680} < \frac{48}{125}.$$

The first expression increases for $s \ge 1$, as its logarithmic derivative is $2/s + 2/(2s+1) - 2/(s+1) > 0$. Thus $\eta_n < 8/875$.

For completeness we rederive the Gaussian estimate used at the radius. The heat identity $\partial_v F_v = F''_v/2$ permits spatial expansion of $F_v(H-t)$ to order four, variance expansion of F_{v+1} to order two, and variance expansion of F'_{v+1}, F''_{v+1} to order one. The cubic coefficient is $(-t^3+3t)/6$, and the quartic coefficient is $(t^4-6t^2+3)/24$. Therefore, for $t = \pm u$, $z_v = H/\sqrt{v}$,

$$H_n(H, \pm u)/s^3 = \pm A(1-z_v^2)\phi(z_v) + B\phi'''(z_v) + \mathcal{E}, \tag{53}$$

where $A = (u^3-3u)/6$, $H_4 = u^4 - 6u^2 + 3$ and $B = H_4/(24\sqrt{v})$. The four error bounds after division by s^3 are respectively

$$\frac{u^5\|\phi^{(4)}\|_\infty}{120v}, \quad \frac{\|\phi^{(5)}\|_\infty}{48v^{3/2}}, \quad \frac{u\|\phi^{(4)}\|_\infty}{8v}, \quad \frac{|u^2-1|\|\phi^{(5)}\|_\infty}{16v^{3/2}}.$$

Gaussian Fourier inversion gives norms at most $3\phi(0) < 6/5$ and $8/\pi < 51/20$. Since $v \ge 41$ and $\sqrt{v} > 6$, for $u \ge 1$ this gives

$$|\mathcal{E}| < E(u) := u^5/4100 + 3u/820 + 17(3u^2-2)/78720.$$

These errors are uniform in H; they use no higher moment of P.

If $A, B > 0$ and $B \le A/4$, the leading part of (53) is at most $\phi(0)(A + 9B^2/(4A))$ for both signs. Indeed, for the positive sign and $|z_v| \le 1$, it is at most $\phi(0)(A - Az_v^2 + 3B|z_v|)$, and complete the square. Outside that interval the first term is nonpositive. For the negative sign the first term is at most $2e^{-3/2}A\phi(0) < A\phi(0)/2$. In both cases use $\|\phi'''\|_\infty < 2\phi(0)$: on $[0,\sqrt{3}]$ the polynomial $3x - x^3$ is at most two, and beyond $\sqrt{3}$ use $x^3e^{-x^2/2} \le 3\sqrt{3}e^{-3/2} < 2$.

For frozen r, b above define

$$Q(u) = -u^3 + \tfrac{3}{2}ru^2 + 3bu + (8/875)r,$$
$$P(u) = \tfrac{2}{5}S^3\left(A^2 + \frac{H_4^2}{256\cdot 41}\right) + S^3E(u)A + \frac{819}{2000}Q(u)A.$$

At $R = 96/25 < 6$ we have $A > 0$, $0 < H_4 < 36A$, hence $B < A/4$. The sign $Q(R) < 0$ is checked before replacing c by its lower endpoint. The upper bound in (52), multiplied by A, is then strictly below $P(R)$, and exact arithmetic gives

$$-P(R) = \frac{2214258283536204102723}{10250000000000000000000} > \frac{1}{5}.$$

To cover each entire exterior ray, differentiation of the exact Gaussian expression gives

$$|\partial_t H_n(H,t)| < \tfrac{2}{5}(t^2/2 + 1/2 + \eta_n).$$

For verification put $y = (H-t)/\sqrt{n}$, $z = H/\sqrt{n}$; the derivative is $n\phi(z) + Ht\phi(z) - ns\phi(sy)$. The second-order Taylor error between $\phi(y)$ and $\phi(z)$ costs $\phi(0)t^2/2$, while $\partial_s[s\phi(sy)] = -\phi''(sy)$ costs $\phi(0)n(s-1)$. Define $L(u) = \frac{2}{5}(u^2/2 + 1/2 + 8/875) + (819/2000)Q'(u)$. At R, the six numbers Q, Q', Q'', L, L', L'' are strictly negative. Since $Q''' = -6$ and $L'' = 2/5 - 6(819/2000) < 0$, all their required signs persist for $u \geq R$. Thus both radial upper bounds decrease from a strictly negative boundary value. The contact inequality excludes every support point with $|t| \geq R$. □

D.2 Higher moments and an actual Fourier event

Higher moments are used only after Theorem D.1. For $x = |X|$ the nonnegative measure $d\mu_0 = (x-1)^2(x+1/2)\,dP$ has mass $\rho - 1$. Direct expansion gives

$$M_4 - 1 = \int \frac{(x+1)^2}{x+1/2}\,d\mu_0,$$
$$M_5 - 1 = \int \frac{x^3 + 2x^2 + 3x + 3/2}{x+1/2}\,d\mu_0,$$

where $M_j = \mathbb{E}|X|^j$. Both ratios increase for $x \geq 0$: they equal $x + 3/2 + 1/[4(x+1/2)]$ and $x^2 + 3x/2 + 9/4 + 3/[8(x+1/2)]$, whose derivatives are nonnegative. Writing their values at $R = 96/25$ as $k_4(R), k_5(R)$, we obtain

$$M_4 \leq \min(R\rho, 1 + k_4(R)(\rho-1)), \quad M_5 \leq \min(RM_4, 1 + k_5(R)(\rho-1)), \quad M_6 \leq RM_5. \tag{54}$$

Also $\gamma^2 \leq M_4 - 1$, by expanding $\mathbb{E}(X^2 - 1 - \gamma X)^2 \geq 0$.

There is no need to assume a central support gap before the first Fourier bound. For independent copies let $E = \{(X - X')^2 \geq 9/25\}$ and let M be any valid fourth-moment ceiling. Then

$$\mathbb{P}(E) \geq \alpha := \frac{(2 - 9/25)^2}{2M + 6} > 0, \quad |X - X'| \geq 3/5 \text{ on } E, \quad \mathbb{E}[(X - X')^2 \mid E] \leq 2/\alpha. \tag{55}$$

Indeed $\mathbb{E}[(X-X')^2; E] \geq 2 - 9/25$ and $\mathbb{E}(X - X')^4 = 2M_4 + 6 \leq 2M + 6$, so Cauchy proves the mass bound. The conditional second-moment bound follows from its unconditional value two. This is an event under one actual iid marginal, not a replacement characteristic function.

The same support proof will be used again on a smaller closed moment cell. Freeze upper bounds r, b for $\rho, |\beta|$ in its polynomial Q, and test a proposed radius before changing any moment bound. For $R \in [5/2, 96/25]$, the gates $A > 0$, $0 < H_4 < 36A$, $Q(R) < 0$, $P(R) < 0$, and the six negative ray signs suffice. Their strictness excludes the boundary atoms as well as the open exterior rays. A radius that fails a gate is not used. The selected finite search is $R = 2.50, 2.51, \ldots, 3.84$; it only chooses among proved implications.

D.3 The Fourier enclosure used by every pass

Let $f(t) = \mathbb{E}e^{itX}$ and $e_n(t) = e^{-nt^2/2}(1 - in\gamma t^3/6)$. For the remainder in (15), the compensated inversion proved in Appendix A gives

$$|E_n| \leq \frac{2\sqrt{n}}{\pi}\left\{\int_0^{1/2} \frac{|\Im[e^{-itH}(f(t)^n - e_n(t))]|}{t^3}dt + \int_{1/2}^{\infty} \frac{|f(t)^n - e_n(t)|}{t^3}dt\right\}.$$

The comparator is signed, has the correct mass and first moment, and its entire high-frequency tail is included. This formula applies at atoms and for arbitrary real H.

For clarity, the exact input and output of the selected Fourier step are as follows. Acquire $|X| \le R$ first, and choose common ceilings $M \le 11.1$, $g \le 1.85$, M_6 for $M_4, |\gamma|, \mathbb{E}X^6$, together with

$$|(3 - M_4)/24| \le A_4, \qquad |(10\gamma - \mathbb{E}X^5)/120| \le Z_5.$$

Put

$$\begin{aligned} E_6 &= M_6/720 + g^2/72 + (1/2 + g^2/144)^3/5, \\ B_6 &= A_4 + Z_5/2 + E_6/4, \\ a_0 = (N-1)/(2N), \quad b_0 &= (45 - M)(1 - 2/N)/96, \\ c_0 \ge \sqrt{\pi/2}(N/(N-1))^{3/2}, \qquad c_1 &\ge c_0/2. \end{aligned}$$

A complete admissible low-frequency triple is

$$\begin{aligned} L_0 &= c_0 Z_5 + c_1 A_4 g + 251 g^3/17280, \\ L_H &= c_0 A_4 + 251 g^2/4800, \\ L_2 &= E_6/(2a_0^2) + B_6^2/(2b_0^3). \end{aligned}$$

One may take $A_4 = \max(2, M-3)/24$ and $M_6 = RM_5$. The sharper fifth-order coefficient used by the finite cover is proved in Appendix A; it follows from the exact identity for $X^2 - \gamma X - 1$, not from an assumed fifth moment.

The high-frequency input is an actual event for two iid copies:

$$\mathbb{P}(\mathcal{E}) \ge \alpha > 0, \qquad |X - X'| \ge d_E > 0 \text{ on } \mathcal{E}, \qquad \mathbb{E}[(X - X')^2 \mid \mathcal{E}] \le M_E.$$

Proposition A.3 supplies a complete six-term bound K_H at $N = 42$, subject to its explicit positive damping and all-index gates. The first event is (55); an opposite-sign event is used only after an actual central gap is acquired below. Different valid windows may be minimized only after all six terms, including the comparator, are present. No characteristic function of an artificial binary law is substituted for f.

The exact scalar output is

$$|E_n| \le K_F + s_F|H|, \qquad K_F = \frac{16}{25}(L_0 + K_H + L_2/\sqrt{N}), \quad s_F = \frac{16}{25}L_H. \tag{56}$$

All square roots and exponential bounds in the finite evaluation are directed outward. The proof in the appendix applies to every actual $n \ge N$ and allows the law and threshold to vary with n. Thus no fixed-law limit is being used as a uniform bound.

D.4 Coarse coefficients before a central gap is known

Write the non-Fourier constant charge in (15) as

$$A_c^{\mathrm{law}} = \frac{3}{2}\beta(J + \gamma) + 3(\rho J_1 - J_2) + \frac{\rho}{2}M_4 - \frac{1}{4}M_5 - \frac{\rho}{2}.$$

Here A_c^{law} does not depend on c; the subscript records its role as the coefficient of c. We will bound it from above, retaining all signs in the later passes.

The following gap inequalities need no gap around zero:

$$-M_4/6 \le J_1 \le M_4/2, \qquad -3J_2 - M_5/4 \le 0, \qquad J \le \rho + 1. \tag{57}$$

For a gap (u, v) of length h, the first pair follows from

$$\int_u^v (t-u)t\,dt + \frac{1}{2}\int_u^v t^2 dt = hv^2/2, \quad \frac{3}{2}\int_u^v t^2 dt - \int_u^v (t-u)t\,dt = h(2u+v)^2/6.$$

Sum with the nonnegative gap weights and use $\int t^2\tau = M_4/3$. On a negative gap, reflection and

$$\frac{1}{3}\int_u^v x^3 dx - \int_u^v (v-x)x^2 dx = u^3(v-u)/3 \geq 0 \quad (0 \leq u \leq v)$$

prove the second assertion; the positive portion only improves it. The last bound follows from $J \leq \rho + 2m$ and $2m \leq 1$.

For $1 \leq r \leq \rho$ and $R \leq 6$, choose $\Gamma(R, r) \geq 0$ with

$$\Gamma(R,r) \geq \max_{r \leq v \leq R} \left\{ -\frac{v^2}{4} + \frac{(R+r)v}{3} - \frac{Rr}{2} + \frac{r^3(2R-r)}{12v^2} \right\}_+ . \tag{58}$$

Then

$$3(\rho J_1 - J_2) \leq \frac{3}{2}\Gamma(R,r)(\rho - \gamma) + \frac{3}{8}\rho^2 J. \tag{59}$$

On the negative portion of a gap discard $|t| < \rho$, where its charge is nonpositive, and put $x = -t$. Since $1 \leq r \leq \rho$ and $|X| \leq R$, the remaining disjoint contributions are bounded by

$$\begin{aligned} 3\int_r^R \tau(-x)(R-x)x(x-r)\,dx &= 3\mathbb{E}\left[X_- \mathbf{1}_{\{X_- > r\}} \int_r^{X_-} (R-x)x(x-r)\,dx \right] \\ &\leq 3\Gamma(R,r)\mathbb{E}X_-^3. \end{aligned}$$

Nonnegativity permits the countable gap sum and these integrals to be interchanged by Tonelli. For $v \in [r, R]$, the inner integral divided by v^2 is exactly the expression in (58). Since $\mathbb{E}X_-^3 = (\rho - \gamma)/2$, this gives the first term of (59). On the positive portion, $t(\rho - t) \leq \rho^2/4$ and $\sum \tau_{u,v} \int_u^v (t-u)dt = J/2$ give the other term. This argument includes gaps crossing zero and atoms at either endpoint.

The finite bound for Γ partitions $[r, R]$ into 64 closed intervals. On each, maximize the concave quadratic at its clipped vertex $2(R+r)/3$ and its nonnegative reciprocal term at the left endpoint. The alternative ceiling $5/2$ follows at $r = 1, R = 6$ from $3v^4 - 28v^3 + 66v^2 - 11 > 0$ on $[1, 6]$: the endpoint values are $30, 205$ and the two interior critical values are $(85 \mp 35\sqrt{5})/2 > 0$. The positive integral representation is increasing in R and decreasing in r, so that endpoint choice encloses the present domain.

The elementary square

$$V = \mathbb{E}[|X|^3(|X| - \rho)^2] = M_5 - 2\rho M_4 + \rho^3 \geq 0$$

gives $\rho M_4/2 - M_5/4 \leq \rho^3/4$. Thus on $\rho \in [l, u]$, $|\beta| \leq b$, $M_4 \leq M$, and with $0 \leq \rho - \gamma \leq \Delta$, two complete constant bounds are

$$\begin{aligned} A_1(\Delta) &= \tfrac{3}{2}b(2u+1) + \tfrac{3}{2}\Gamma(R,l)\Delta + \tfrac{3}{8}u^2(u+1) + u^3/4 - l/2, \\ A_2 &= \tfrac{3}{2}b(2u+1) + 2uM - l/2. \end{aligned} \tag{60}$$

Use $A = \min(A_1, A_2)$. The first uses (59); the second uses (57). They are alternative complete bounds, not partial costs to be mixed.

For the linear term K_1, the following three bounds are valid:

$$\begin{aligned} K_{\text{rad}} &= 3ub + \frac{R(R+2-3l/2)}{R+1/2}(u-1), \\ K_{\text{univ}} &= 6u - 3u^2/2 + 3b, \\ K_R &= Ru/2 + \max_{r \in \{l,u\}} |R/2 - 3r/2|g + 3b. \end{aligned}$$

For the first use the positive measure $d\mu_0 = (|X|-1)^2(|X|+1/2)dP$, of mass $\rho - 1$, and the exact identity

$$K_1 + 3\rho\beta = \int \operatorname{sgn}(X)\frac{x(x+2-3\rho/2)}{x+1/2}\,d\mu_0, \qquad x = |X|.$$

The ratio is convex in $x \geq 0$ and decreases in ρ. Its negative part is at most $31/40$ on $\rho \leq 1.85$; require its displayed endpoint ceiling to exceed that number. The second bound uses $|\mathbb{E}[X|X|^3] - 3\gamma| \leq 3\rho$ for $R \leq 6$, and $\rho \leq 2$. For the third use the signwise fourth-moment bounds $0 \leq \mathbb{E}X_\pm^4 \leq R(\rho \pm \gamma)/2$. Finally (57) gives

$$|\phi_z\Omega| \leq v_H := \frac{2}{5}\max\{1/2, 2M/3 - 1/2\}, \qquad \Omega = J_1 + (M_4 - 3)/6.$$

Lemma D.2 (A stronger signed fourth-moment charge)**.** *Let $|X| \leq R$, $\rho \in [l,u]$, $|\beta| \leq b < 1$. Put $h_* = 1 + b$ and $a(r) = (3r - R)/2$. A valid bound for $|K_1|$ is the maximum on $[l,u]$ of*

$$B_R(r,b) = Rr - \tfrac{3}{2}r^2 + 3b + \tfrac{1}{2}(1+b)a(r)^2.$$

If the three gates

$$h_* \geq a(u), \qquad 3 + 2a(l)h_* - \tfrac{3}{2}h_*^2 \geq 0, \qquad R - 3u + \tfrac{3}{2}h_*^2 \geq 0$$

hold, it may be intersected with the smaller bound

$$Ru - \tfrac{3}{2}u^2 - 3 + 3h_* + a(u)h_*^2 - h_*^3/2.$$

Proof. For $\epsilon = \pm 1$ set $h = 1 - \epsilon\beta > 0$ and $\delta = \rho - \epsilon\gamma$. Signwise Cauchy and $m \leq 1/2$ give $\delta \geq h^2$. The fourth moments on the two signs give

$$\epsilon K_1 \leq R\rho - \tfrac{3}{2}\rho^2 - 3 + 3h + a(\rho)\delta - \delta^2/(2h).$$

Completing the square in unrestricted δ and using $h \leq h_*$ proves the first bound. With $\delta \geq h^2$, the last three terms maximize to

$$\psi_a(h) = \begin{cases} h(3 + a^2/2), & 0 < h \leq a, \\ 3h + ah^2 - h^3/2, & h \geq a. \end{cases}$$

The first branch is increasing. The second has derivative $3 + 2ah - 3h^2/2$, a concave quadratic whose endpoint values on $[0, h_*]$ are nonnegative under the second gate. The branches agree at $h = a$; the first gate places h_* in the second branch. The remaining derivative in ρ is $R - 3\rho + 3h_*^2/2$, nonnegative by the third gate. This proves the claim, including all endpoints. The quadratic maximum in the first bound is evaluated at its endpoints and, when present, its concave vertex. □

Take the minimum of the applicable bounds for $|K_1|$ and call it K. No assertion about the signs of $\beta\gamma$ or $K_1\gamma$ is used.

D.5 Two quadratic passes and the first actual gap

Use the 24 closed cells with endpoints

$$r_i = 1 + 3361i/96000, \quad i = 0, \ldots, 24, \qquad N = 42,$$

and directed constants

$$c_- = 0.4097321837023 < C_{\mathrm{E}} < c_+ = 0.4097321837025,$$
$$\phi_l = 997/2500 < \phi(0), \quad \sigma_l = 3.162277660 < \sqrt{10}.$$

The Gaussian constants follow from the rational Machin estimates in Lemma B.1 and Appendix A; the narrow c enclosure has the explicit directed recipe in Lemma B.2. The interval encloses the fixed coefficient c, not the actual normalized score.

On a cell $[l, u]$, form the moments and the universal actual event above, and compute the complete Fourier pair K_F, s_F. In pass 1 use $\Delta = 2u$, $A = \min(A_1(2u), A_2)$, and, for $t \in [c_-, c_+]$,

$$f(t) = tK + v_H + s_F, \qquad q(t) = tl - g/5.$$

Require $q > 0$. The bound $\mathcal{R}_c \le K_F + cA + f(c)|H| - q(c)H^2$ is affine in c at each fixed $|H|$, so its maximum is at an endpoint before completing squares. Let an overline denote upward rounding to 10^{-10}. Compute

$$\begin{aligned}
T_1 &= \overline{\max_{t=c_-,c_+} \{K_F + tA + f(t)^2/(4q(t))\}},\\
\eta_1 &= (c_- l - T_1/N)/2,\\
U_1 &= \min\left\{T_1, \overline{\max_{t=c_-,c_+} \{K_F + tA + f(t)^2/[4(q(t)+\eta_1)]\}}\right\},\\
d_1 &= \overline{6U_1/(N\phi_l)}.
\end{aligned} \tag{61}$$

The strict gate $0 \le T_1 < Nc_- l$ is checked before η_1 or the resource bound is used. Corollary 4.2 then proves $\mathcal{R}_c \le T_1$, $\mathcal{U} \le U_1$, and $nD \le Nd_1$. The scalar bounds hold at every actual $n \ge N$.

For the second pass the circle in Lemma 3.1 gives

$$D \ge \sigma\rho - \gamma - 3\sqrt{\rho^2 - \gamma^2}, \qquad \sigma = \sqrt{10}.$$

Put $a_0 = \sigma_l - d_1/l$. A valid lower bound for γ/ρ is

$$Y(a_0) = \begin{cases} -1, & a_0 \le -1,\\ (a_0 - 3\sqrt{10 - a_0^2})/10, & -1 < a_0 \le \sqrt{10}. \end{cases} \tag{62}$$

Indeed $y + 3\sqrt{1-y^2}$ increases from -1 to $\sqrt{10}$ on $[-1, 1/\sqrt{10}]$. Solving its level equation on that branch gives the smaller quadratic root in (62); it has $a_0 - y \ge 0$, so squaring introduced no false branch. Every y below it violates the required lower level. A downward rational bound y_l is obtained by rounding the square root upward. Thus

$$\begin{gathered}
g_l = \max\{-g, \sigma_l l - 3 - d_1, \min(ly_l, uy_l)\} \le \gamma,\\
\Delta = \min\{2u, (1-\sigma_l)l + 3 + d_1, u - g_l\}, \qquad \phi_- = \phi_l(1 - T_1/(Nc_- l)) > 0,\\
A = \min\{A_1(2u), A_1(\Delta), A_2\},\\
q(t) = \max\{tl - g/5, tl + \tfrac{1}{2}\min(\phi_- g_l, (2/5)g_l)\}.
\end{gathered}$$

The first two bounds also use $2m \le 1$. Each value of $q(t)$ is affine in t, since its two alternatives have the same slope l. Repeat the raw bound first, intersect it with T_1, then price phase using this new raw bound, intersect the resource with U_1, and finally retain $d_2 \le d_1$. A resource bound is never substituted for a raw bound in the density or phase license.

Certificate condition D.3 (The two coarse passes)**.** The preceding coefficient domains, Fourier gates, endpoint enclosures and strict raw gates hold on every one of the 24 closed cells. Outward evaluation gives the following ceilings, and in every second-pass cell $d_2 < (\sigma_l - 1)l$:

pass	$\max T$	$\max U$	$\max d$	central gap proved
1	13.2818901598	10.9330610254	3.9164138937	23
2	11.5164967977	9.9790315907	3.5746638454	24.

The last column counts the cells satisfying $d_j < (\sigma_l - 1)l$, which excludes zero from the support by the argument below. The minimum second-pass gap margin is at least $526024602543/1600000000000 > 0$.

The stated condition is a finite exact arithmetic obligation. The named two-pass proof object records and rechecks these inequalities; it is kept distinct from the analytic implications just proved. Although its implementation retains additional previously valid minima, the fresh quadratic ceilings attain all the displayed minima in this cover, so no earlier resource-refinement schedule is a mathematical premise here.

If zero belonged to the closed support, every gap would be on one side of zero and $J \le \rho$. Hence $D \ge (\sigma-1)\rho$, contradicting the strict second-pass test. Centering and positive variance therefore give an actual central gap $(-A, B)$, with positive endpoints. This acquires an event, rather than supposing an artificial two-point kernel.

D.6 Signed moments and physical gap costs

Write $\mu_4 = M_4$, $\eta_5 = M_5$, $\sigma = \sqrt{10}$. Put $a_\pm = (1 \pm \beta)/2$, $b_\pm = (\rho \pm \gamma)/2$. Both signs have positive probability and positive first moment, since a centered variance-one law cannot be supported on one half-line. Thus $|\beta| < 1$ and the signwise denominators below are positive whenever used; degenerate correction denominators are handled separately. Cauchy gives $m^2 \le a_+ a_-$ and $b_\pm m \ge a_\pm^2$. Consequently

$$\frac{(1+\beta)^{3/2}}{\sqrt{1-\beta}} - \rho \le \gamma \le \rho - \frac{(1-\beta)^{3/2}}{\sqrt{1+\beta}}, \qquad m \ge \max\left\{\frac{1+\beta^2}{2\rho}, \frac{a_+^2}{b_+}, \frac{a_-^2}{b_-}\right\}.$$

These statements also hold with a zero atom. Projection of $X|X|$ onto the span of X and $|X|$ gives

$$\mu_4 \ge \rho^2 + \frac{(\gamma - \rho\beta)^2}{1-\beta^2}.$$

For completeness, this follows by minimizing the nonnegative expectation $\mathbb{E}(X|X| - uX - v|X|)^2$ over u, v; its Gram matrix has diagonal entries one and off-diagonal entry β, and is positive definite.

Two support-localizing moment matrices give the additional upper bounds as follows. If $D_\pm = Ra_\pm - b_\pm > 0$, then

$$2m + \sum_\pm \frac{(Rm - a_\pm)^2}{D_\pm} \le R, \qquad \mu_4 \le R\rho - \sum_\pm \frac{D_\pm^2}{Rm - a_\pm}.$$

Indeed the first follows by summing the determinant inequalities for

$$\mathbb{E}[(R - |X|)(1, |X|)^{\mathsf{T}}(1, |X|); \pm X > 0].$$

The second follows from the corresponding matrices with an additional factor $|X|$. One more factor $|X|$ yields, sign by sign,

$$\mathbb{E}[X_\pm^5] \le R\mathbb{E}[X_\pm^4] - \frac{(Rb_\pm - \mathbb{E}[X_\pm^4])^2}{D_\pm}.$$

Zero denominators are treated by dropping their nonnegative correction; they never imply exclusion without a separate contradictory numerator. The common m is retained throughout.

A second useful upper bound for m follows by integrating $(x-a)^2(R-x) \geq 0$ for $x = |X| \in [0, R]$. For every fixed $a > 0$,

$$2m \leq \frac{R - \rho + 2a + Ra^2}{a(2R + a)}.$$

The cover uses only the eleven rational choices $a = 2/10, \ldots, 12/10$.

Write $\delta = \gamma - \rho\beta$. Cauchy and a square completion on each sign give the signed constant bound

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\eta_5 \leq \frac{\rho^3}{4} - \frac{(\rho+\gamma)(\delta_+)^2}{8(1+\beta)^2} - \frac{(\rho-\gamma)((-\delta)_+)^2}{8(1-\beta)^2}.$$

Indeed on either sign, write $a = \mathbb{E}X_s^2$, $b = \mathbb{E}X_s^3$, $c = \mathbb{E}X_s^4$, $f = \mathbb{E}X_s^5$. Cauchy gives $c \geq b^2/a$, $f \geq c^2/b$, and $\rho c/2 - f/4 \leq \rho^2 b/4 - (c - \rho b)^2/(4b)$. Distance to the half-line $c \geq b^2/a$ proves the stated penalty on summing both signs. Independently, $\mu_4^2 \leq \rho\eta_5$ and the projection bound above give

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\eta_5 \leq \frac{\rho^3}{4} - \frac{(\gamma - \rho\beta)^4}{4\rho(1-\beta^2)^2}.$$

Each is a complete deduction from the same square. There is a third independent deduction which is often stronger. For $Y = |X|$ and $\delta = \gamma - \rho\beta$, put

$$V = \mathbb{E}[Y(Y^2 - \rho Y)^2] = \eta_5 - 2\rho\mu_4 + \rho^3.$$

Since $\delta = \mathbb{E}[Y \operatorname{sgn}(X)(Y^2 - \rho Y)]$, weighted Cauchy gives $\delta^2 \leq 2mV$. Hence

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\eta_5 \leq \frac{\rho^3}{4} - \frac{\delta^2}{8m}. \tag{63}$$

The maximum of these three separately valid losses can be subtracted. Their sum cannot. A zero atom contributes zero to the weighted measure.

An actual central gap without assuming $D < 2$

The physical decomposition (12) shows that $D < (\sigma - 1)\rho$ implies an actual central gap. Use its same nonnegative terms, writing $T = T_+ + T_-$ and $Q = (\sqrt{\sigma+1}A - \sqrt{\sigma-1}B)^2$. Hence $T \leq D/(\sigma - 1)$. The additional inequality $J \leq \rho + 2mAB$ gives $6mAB \geq (\sigma-1)\rho - D$. Consequently, if $\rho \geq l$, $D \leq d < (\sigma_l - 1)l$ and $m \leq m_u$,

$$(A+B)^2 \geq \frac{l - d/(\sigma_l - 1) + \{(\sigma_l - 1)l - d\}/3}{m_u} > 0. \tag{64}$$

This is the distance lower bound used in the complete high-frequency event. It uses the actual moment m and the actual support endpoints.

Moreover $A \leq a_-/m$, $B \leq a_+/m$ and $J \leq \rho + 2m$. The last bound gives the compulsory defect

$$D \geq \sigma\rho - \gamma - 6m.$$

The inequalities $T - J_o \geq 2BR_+ + 2AR_-$ give the useful additional endpoint constraint

$$D \geq (\sigma - 1)b_+ + 6a_+B - (\sigma + 5)mB^2.$$

If $D < (\sigma - 1)b_+$, this downward quadratic forces B to be at least its positive root. The reflected constraint has a_-, b_-, A and the coefficients $\sigma + 1, \sigma + 7$. In a closed moment box,

positive coefficients are rounded down, negative coefficients up, and the lower square-root endpoint is used in the resulting root bound.

The actual opposite-sign event has probability $2pq$. There is no atom at zero. Cauchy gives the necessary probability interval $2m_-^2/(1+\beta_+) \le p \le 1 - 2m_-^2/(1-\beta_-)$ on a moment cell. In particular, $2pq \ge (1+\beta^2)^2/(2\rho^2)$, using $m \ge (1+\beta^2)/(2\rho)$ and $pq \ge m^2$. The minimum of $p(1-p)$ on a closed interval is attained at an endpoint.

For independent copies one has exactly

$$\mathbb{E}[(X-X')^2 \mid XX' < 0] = \frac{1+\beta(1-2p)+4m^2}{2pq} = \frac{1+4m^2+\beta}{2p} + \frac{1+4m^2-\beta}{2q}.$$

For an upper bound, use the upper m endpoint and both endpoints of p and β. The expression is convex in p and affine in β. This improves the former $2/\mathbb{P}(XX' < 0)$ bound inside the complete high-frequency estimate. Both copies still have the same actual marginal.

Preserving central and outer integral costs

Let

$$P_1(A,B) = B^3/3 + AB^2/2 - A^3/6, \quad P_2(A,B) = B^4/4 + AB^3/3 - A^4/12, \quad G = \rho P_1 - P_2.$$

Then $J_{1,c} = mP_1$, $J_{2,c} = mP_2$. Since $\partial_B G = B(A+B)(\rho - B)$,

$$G(A,B;\rho) \le G(A,\rho;\rho) = \rho^4/12 + \rho^3 A/6 - \rho A^3/6 + A^4/12.$$

The last expression increases with A because its derivative is $(\rho - A)^2(\rho + 2A)/6$, and is convex in ρ because its second ρ derivative is $\rho^2 + \rho A$. Its closed-rectangle maximum is therefore bounded using the upper A endpoint and both ρ endpoints.

Positive outer gaps satisfy $(t-a)(\rho - t) \le (\rho - B)_+^2/4$. On negative outer gaps the positive part is bounded by $(R-\rho)^2/4$ after extracting $|t|$. Integration gives

$$3(\rho J_1 - J_2)_o \le \frac{3}{8}\{(\rho - B)_+^2 T_+ + (R-\rho)^2 T_-\}.$$

The available weighted outer budget is $D - mQ$, not the entire D. The cover intersects this budget with the individual bounds $T_+ \le b_{+,u} - m_l B_l^2$ and $T_- \le b_{-,u} - m_l A_l^2$.

For the signed phase term $\Omega = J_1 + (\mu_4 - 3)/6$, retain

$$-RT_-/2 \le J_{1,o} \le RT_+/2.$$

The central minimum of P_1 occurs at B_l and an A endpoint; its maximum occurs at B_u and A clipped to the interval at $A = B_u$. A nonnegative polynomial bound is multiplied by m_u for an upper bound and m_l for a lower bound; a negative one uses the opposite endpoint. For example, the binary endpoints $A = 3/2$, $B = 2/3$ make this maximum negative; the sign switch is necessary even when $D < 2$. The fourth-moment sign intervals also sharpen $K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2$. The complete K_1, Ω, ϕ_z rectangles are used in the signed expression $|cK_1 + \phi_z \Omega|$, without replacing it by two separate absolute values.

A compulsory first moment before each signed evaluation.

The physical inequality $D \ge \sigma\rho - \gamma - 6m$ is especially useful inside a cell. If its prior ceiling is $D \le d$, and $\rho \ge l$, $\gamma \le \gamma_+$, then

$$m \ge m_* := \max\{0, (\sigma_l l - \gamma_+ - d)/6\}. \tag{65}$$

If $m_* > 1/2$, the cell is empty. Otherwise the common sign-moment inequality $4m^2 + \beta^2 \le 1$ forces $|\beta| \le \sqrt{1 - 4m_*^2}$. This restriction must be intersected with the beta interval for this

gamma cell, not carried from a different gamma cell. Recompute all dependent sign-moment ranges after the intersection. These are necessary conditions on one law; no sufficiency or moment-realizability assertion is made.

For completeness, the event distance and conditional moment refinements fit the Fourier input exactly. On the acquired actual gap, use $\mathcal{E} = \{XX' < 0\}$, a downward square root of (64), and the larger of the two valid probability lower bounds

$$\alpha = \max\left\{\frac{(1+\min|\beta|^2)^2}{2u^2},\quad 2\min_{p=p_-,p_+} p(1-p)\right\}.$$

The conditional second-moment ceiling is the smaller of $2/\alpha$ and the corner maximum of the exact expression displayed above. Its numerators are positive, it increases with m, is affine in beta, and is convex in p, so these corner choices are valid. Each event option is inserted into all six HIGH terms. Taking the minimum with the complete universal-event bound preserves the same actual Fourier integral. None of these statements follows merely from declaring a moment rectangle feasible.

D.7 The signed enclosure and its ordered update

On one closed moment cell retain bounds for (ρ, β, γ, m), an already proved support radius R, an incoming raw ceiling T, and the weighted defect invariant $nD \le Nd$. Thus $D \le d$ for every actual $n \ge N$. Intersect all necessary moment conditions just proved, with directed endpoint operations. A positive numerator with a zero denominator proves incompatibility only through its original positive-semidefinite inequality; a zero-over-zero correction is dropped. No division is made before this check. Bisection of the monotone first-moment necessary inequality keeps its upper endpoint, so its rounding cannot remove a law.

Here is an explicit description of the coefficient enclosures. Let M be the resulting upper fourth moment and let $[J_-, J_+]$ contain J; one can use

$$J_- = \max\{0, ((3+\sigma_l)l - \gamma_+ - d)/3\},\quad J_+ = \min\{u + 2m_+, ((3+\sigma_u)u - \gamma_-)/3\},$$

with a rational $\sigma_u > \sqrt{10}$. For the beta term take the full corner maximum of $3\beta(J+\gamma)/2$. For the gap charge take the minimum of the complete coarse charge (59) and the central-plus-outer charge just proved. Write this bound as G_* and the beta-term maximum as B_*. Let $P_* \ge 0$ be any lower bound for the maximum of the three independent fifth-moment penalties. Then a valid complete constant bound is

$$A = \min\{B_* + G_* + u^3/4 - l/2 - P_*,\quad B_* + 2uM - l/2\}. \tag{66}$$

The fourth semimoment inequalities and the signed product rectangle for $\rho\gamma$ give an interval $[k_-, k_+]$ containing $K_1 = \mathbb{E}X_+^4 - \mathbb{E}X_-^4 - 3\beta - 3\rho\gamma/2$. The central polynomial mP_1 and the signed outer bounds give an interval $[\omega_-, \omega_+]$ containing $\Omega = J_1 + (M_4 - 3)/6$. Intersect them with every previous valid interval. When multiplying by a moment endpoint, the sign of the coefficient is retained. These constructions explain every coefficient in the next quadratic, without assuming that unrelated corner maxima are attained by one common law.

Require the incoming raw gate $0 \le T < Nc_-l$. By Corollary 4.2,

$$\phi_z = \phi(H/\sqrt{n}) \in [\phi_-, 2/5], \qquad \phi_- = \phi_l(1 - T/(Nc_-l)) > 0.$$

For a formal coefficient $v \in [c_-, c_+]$ set

$$\begin{aligned} B_v &= K_F + vA, \\ F_v &= \max_{k\in\{k_-,k_+\},\ \omega\in\{\omega_-,\omega_+\},\ \psi\in\{\phi_-,2/5\}} |vk + \psi\omega| + s_F, \\ Q_v &= vl + \tfrac{1}{2}\min\{\phi_-\gamma_-, (2/5)\gamma_-\}. \end{aligned} \tag{67}$$

The actual raw remainder is at most $B_c + F_c|H| - Q_c H^2$. The maximum over the signed rectangle is valid by successive affine maximization, and the absolute value preserves convexity. For fixed $|H|$, this expression is convex in c: the constant and curvature parts are affine and F_c is convex. Thus the two endpoints c-,c+ suffice, even if A is negative. A negative lower gamma must be priced with the upper density endpoint; it cannot be reflected inside the contact law.

Lemma D.4 (One ordered signed step)**.** *Suppose all preceding cell bounds hold and $Q_v > 0$ at both endpoints. Let upward rounding be applied to every proposed numerical ceiling. Set*

$$\begin{aligned} T' &= \min\{T, \max_{v=c_-,c_+} (B_v + F_v^2/(4Q_v))\}, \\ \eta &= (c_l - T'/N)/2, \\ U' &= \max_{v=c_-,c_+} (B_v + F_v^2/[4(Q_v + \eta)]). \end{aligned}$$

A negative T' or U' excludes the cell. Otherwise $\mathcal{R}_c \le T'$, $\mathcal{U} \le U'$, and the new weighted defect bound is

$$nD \le Nd', \qquad d' = \min\{d, 6U'/(N\phi_l)\}.$$

A valid raw-threshold ceiling is

$$|H| \le \max_{v=c_-,c_+} \frac{F_v + \sqrt{F_v^2 + 4(Q_v + \eta)\max(0, B_v)}}{2(Q_v + \eta)}. \tag{68}$$

Proof. Complete the two raw squares first. The incoming strict gate persists under $T' \le T$, and gives $\eta > 0$. Only now apply Corollary 4.2 and complete the two resource squares with curvature $Q_v + \eta$. A negative complete ceiling is incompatible with the nonnegative raw or resource expression. The resource contains $n\phi(0)D/6$, so the displayed weighted bound follows without setting the actual sample size equal to N. If the maximum of both resource quadratics at a phase were negative, no witness could have that phase. Replacing B by its positive part enlarges each positive root and gives (68). Previous valid resource or phase bounds may also be intersected, but an unlicensed resource bound cannot be fed back as a raw ceiling. □

D.8 Closed coverage and the selected acquisition theorem

For each of the original 24 rho cells, acquire a support radius with the independent heat/ray test before recomputing higher moments. Each signed pass then partitions a full necessary beta enclosure into eight closed intervals, each resulting necessary gamma enclosure into eight closed intervals, and each necessary common-m enclosure into two closed intervals. The original beta interval is restored for every gamma slice before applying (65). Discarding an outer interval counts all its descendants, and requires one of the proved necessary conditions to be inconsistent. In every processed rho cell the active and excluded counts sum to 128.

Each surviving actual law satisfies the cell update of Lemma D.4. Pool upper bounds by maxima and lower bounds by minima before the next pass. This may enlarge the possible set, but cannot lose an actual witness. A rho cell disappears from later passes only when all of its descendants have been excluded. Degenerate closed intervals may duplicate boundary work; they do not leave holes.

For export, intersect each surviving necessary probability interval with all 100 closed panels $[j/100, (j+1)/100]$, $j = 0, \dots, 99$. A cell meeting a shared endpoint contributes to both panels. From $\theta = 1 - m^2/[p(1-p)]$ obtain a panel upper bound using its lower m endpoint and the maximum of $p(1-p)$ on the intersection, including $1/4$ if it crosses $1/2$. Pool this simultaneously with the raw, weighted defect, rho, beta, and phase bounds of the same cells.

Certificate condition D.5 (The four signed passes)**.** Starting with Condition D.3, all support, moment, actual-event, Fourier, denominator, signed-curvature and ordered raw-license gates hold on the complete four-pass cover just described. Its exact rational evaluation contains 11264 nominal signed boxes in total. The complete 100-panel export satisfies, on every nonempty panel,

$$\begin{aligned} \mathcal{R}_c &\le 2.1995174, & nD &\le 32.20665378, & D &\le 0.76682509, \\ \rho &\le 71527/48000, & |\beta| &\le 0.50549963, & |H| &\le 1.83605680. \end{aligned}$$

Fourteen rho cells survive, and thirty probability panels are nonempty. The largest support radius among the surviving final rows is 157/50; thus the export also supplies

$$\operatorname{supp} P \subset (-157/50, 157/50). \tag{69}$$

This is the named fixed-target coarse proof object, with its own targets and complete incidence requirements. Its reproduction is specified in Appendix E; numerical acceptance remains distinct from the analytic proof that its leaves enclose all possible witnesses. The preceding statements supply that reduction and explicitly identify the finite predicates used. They do not change an older certificate's hypotheses or its target.

Proof of Theorem 5.1. Theorem D.1 supplies the actual support before higher moments are used. Appendix A and the actual-event constructions prove the complete Fourier bounds on their specified domains. The coarse coefficient inequalities and the two ordered quadratic steps establish the first actual gap. Every signed cell thereafter uses the moment and physical-gap bounds of this same law, followed by Lemma D.4. Induction over the four passes preserves the weighted invariants. The complete closed cover and outward pooling prove that every witness reaches an incident output panel with the stated bounds. □

The argument contains no upper bound for the witness's actual score, no scalar score recurrence, and no second deleted-score assumption. It uses only the quadratic improvement in the threshold term. Sections 6 and 7 replace downstream panel arguments by one support region and two direct comparisons at $n \ge 700$. The internal entry cover remains intact. The independent finite base is a separate input.

E Certificates and independent reproduction

The analytical hypotheses, the finite numerical verification, and the missing finite base are separate obligations. This appendix identifies the computation used by Theorem 5.1; it does not supply the base assumed in Theorem 8.2.

E.1 What is stored and what is recomputed

The retained format and target are, respectively,

```
FIXED_C_COARSE_ENTRY_V7
INITIAL42_FIXED_TARGET_COARSE_V7.
```

The JSON object stores four lists of 24 moment-cell records and all 100 probability-panel records. A live cell records its support radius, remainder and defect ceilings, and active/excluded descendant counts; a previously excluded cell is marked accordingly. It does not store 11264 individual leaves. The checker reconstructs the subdivisions, exclusions, bounds, and panel incidences. The soundness argument is Proposition 5.3.

For example, the first signed pass records a checkpoint for cell index 0. The checker reconstructs its moment interval as $[1 + 3361i/96000, 1 + 3361(i+1)/96000]$ at index i; the JSON stores i, not these two endpoints. Its descendants and their enclosures must still be recomputed.

Quantity for ρ-cell 0	Value and provenance
Moment interval (reconstructed)	$[1, 99361/96000]$
Support radius (stored)	$5/2$
Remainder ceiling	$27071971/50000000 = 0.54143942$
Defect ceiling d	$18233099/100000000 = 0.18233099$
Active / excluded descendants	128 / 0

The last numerical bound means $nD \le 42d = 7.65790158$ for every actual $n \ge 42$ in the cell, rather than only for a law at index 42.

All paths below are relative to the repository root. The retained source and data can be read at the immutable baseline commit linked in Section 9; they are unchanged by this revision.

Certificate
`mathematics/fixed-target-v7/2026-10-27/entry-proof.json`

Checker
`mathematics/fixed-target-v7/2026-10-27/accept_entry.py`

Source-isolated replay
`mathematics/fixed-target-v8/2026-10-28/isolated_entry_replay.py`

Selected exporter and exploratory producer
`mathematics/fixed-target-v7/2026-10-27/export_entry.py`, `mathematics/fixed-target-v7/2026-10-27/signed_probe.py`

Two-pass ancestor
`mathematics/fixed-target-v6/2026-10-27/signed-two-pass-check.py` and its pinned dependencies.

The exploratory producer considers candidate subdivisions, support radii, and passes. The exporter selects the first four stages of a six-stage discovery trace and rounds ceilings outward. Acceptance reconstructs the two coarse and four signed passes; it does not rely on a discovery success label. The selected path uses integer and rational arithmetic. Generation and checking share mathematical kernels, so this is not a claim of independent implementations.

E.2 The acceptance procedure

The acceptance program carries out the following steps in a fixed order.

1. Check the format and target names, exact record fields, canonical rational strings, four ordered lists of 24 cells, and all 100 ordered closed panels, including their numerical target domains. Check source hashes before importing the numerical kernels.

2. Recompute the two coarse passes. Their 24 endpoint pairs must agree with $[1 + 3361i/96000, 1 + 3361(i+1)/96000]$.

3. For each live cell in each signed pass, recompute the support radius and the $1 \times 8 \times 8 \times 2$ subdivision in (ρ, β, γ, m). Evaluate the exclusions and complete upper bounds on these descendants using the previous pass's ceilings.

4. Require the active and excluded counts to agree with the stored counts and to sum to 128. For a surviving cell, require each stored ceiling to enclose the recomputed value and to be no larger than the previous ceiling. A cell excluded in an earlier pass remains excluded.

5. Reconstruct every final probability-panel incidence, including shared endpoints. Check the stored incidence counts and require every stored panel ceiling to enclose the corresponding recomputed maximum.

6. Report the resulting maxima and run the rejection controls.

Individual descendant boxes are not archived in the object. Thus the comparison is between deterministic recomputation and stored counts and ceilings, not between two stored lists of discarded boxes. The mathematical reason every possible witness is covered is given in Appendix D and Proposition 5.3.

A guide to the numerical sources.

The following is a reading map for the retained implementation. Relative paths beginning with `v5/`, `v6/`, and `v7/` below abbreviate `mathematics/fixed-target-v5/2026-10-27/`, `mathematics/fixed-target-v6/2026-10-27/`, and `mathematics/fixed-target-v7/2026-10-27/`, respectively.

`v7/accept_entry.py`: `shape`, `verify`
Record validation, source pins, ordered replay and output comparisons.

`v6/signed-two-pass-check.py`: `one_pass`, `run`
The two coarse quadratic passes and the first central-gap test in Section D.5.

`v7/signed_support.py`: `support`, `cell_radius`
Support-radius gates from the contact and Gaussian heat estimates in Theorem D.1.

`v7/signed_core.py`: `stage`
Signed subdivisions, moment constraints, gap and remainder charges, ordered resource updates, and final panel incidences.

`v5/cell-sources/kernel.py`: `sixth`, `universal_high`, `exphi`
Low-frequency constants, the universal actual-event bound and directed exponential enclosures. The replacement high-frequency window routine is `install` in `v5/cell-sources/high_window_core.py`; it compares complete six-term bounds from Proposition A.3.

This map is not a separate verification of those programs. An independent checker must establish the same implications for every covered box, including signs, denominator domains and endpoints; copying a success message or using the same kernel twice does not do so.

E.3 Exact arithmetic and its limits

Rationals are represented by integer numerators and denominators. Outward rounding deliberately weakens a bound. Square-root endpoints are checked by rational square comparisons. For a nonnegative numerator, an upper numerator bound divided by a positive lower denominator bound gives an upper quotient bound. Signed quotients require the endpoint extrema: if $a \in [a_-, a_+]$ and $0 < b_- \le b \le b_+$, then

$$\frac{a}{b} \in \left[\min_{i,j\in\{-,+\}} \frac{a_i}{b_j}, \max_{i,j\in\{-,+\}} \frac{a_i}{b_j} \right].$$

A denominator interval containing zero requires a separate argument; this rule does not permit division by it. The exponential helper uses, for rational $0 \le y \le \min(x, 64)$,

$$e^{-x} \le e^{-y} \le \left(\sum_{j=0}^{64} \frac{y^j}{j!} \right)^{-1},$$

followed by upward rounding. The positive exponential series proves this enclosure. The Gaussian constants, including the narrow interval for C_{E}, are proved in Appendix B.

Numerical precision determines whether an enclosure is sharp enough to prove a strict inequality; extra digits alone do not prove it. Uniformity in n comes from the analytical majorants in the Fourier appendix, not from sampling indices or assuming monotonicity of C_n. The checker semantics, their correspondence with the displayed formulas, and the integer/rational implementation remain part of the computational trust boundary. Hashes identify bytes, not mathematical truth. Subdivision certificates have established methodological precedents [15]; that precedent does not validate this custom format or transfer a proof assistant's guarantee to Python.

E.4 Three separate checks

Use a fresh output path for each replay. From the repository root, the retained entry is reproduced by

```
python3 -I -B \
  mathematics/fixed-target-v8/2026-10-28/isolated_entry_replay.py \
  --output-dir runs/compact-v16-entry-review
```

The wrapper copies a fixed 13-file source/data set into an isolated source environment. The wrapper status is `PASS_ISOLATED_REPRODUCTION_FIXED_ENTRY_V7`; its numerical output is `PASS_INITIAL42_FIXED_TARGET_COARSE_V7`. It reconstructs 11264 nominal signed boxes, 100 panels, 30 nonempty panels and 14 surviving moment cells. Nineteen checker rejection controls and five wrapper controls test malformed or altered inputs. They supplement rather than replace the coverage proof. A recorded replay on 29 October 2026, using Python 3.9.6 without optimization, took 67.7 seconds. This is one observed run, not a performance guarantee; its command, source hashes, return code and output are preserved in `publication/entry/receipt.json` and the neighboring captures. The Python standard library suffices for these commands.

The bounded historical checks and the current document's reference structure are reproduced separately by

```
python3 -I -B \
  manuscript/compact-v16/tools/recheck.py \
  compact-v16-bounded-review
```

Here the output path is relative to `manuscript/compact-v16/`; an absolute fresh path with an existing parent can instead be supplied. This runs nine unchanged v16 mathematical programs. They retain their original parameters, including the former small-noise minorant. They are regression evidence for the retained inputs, not verification of the revised scalar choices and not a replay of the coarse entry.

The changed scalar claims have their own exact checks:

```
python3 -I -B \
  manuscript/compact-v16/tools/check_revision_constants.py \
  --output-dir runs/compact-v16-scalar-review
```

This runs the geometry, count, and foundation checks. They reconstruct the rational comparisons from their definitions: the endpoint probability exclusion, six ray bounds, the new minorant's full rectangle argument, the common error envelope and its two quartic endpoints, the Fourier quadratic identity, and the narrow C_{E} enclosure. The continuous proofs of sign, concavity, monotonicity, and coverage remain in the text. Selected atomic-law examples are diagnostics, not an all-law proof. The revision record reports these executions separately from the inherited entry replay and from document compilation.

References

[1] Logan Bell. Extremizers for the independent Berry–Esseen inequality. Manuscript dated 2026-10-06; collection version 0.7.0; source snapshot f61bc638774bfa1165065462f98fe31efade30bf, 2026. URL https://github.com/epicycloids/independent-berry-esseen-bounds/blob/f61bc638774bfa1165065462f98fe31efade30bf/paper/extremizers/extremizers.pdf.

[2] Logan Bell. An upper bound on the Berry–Esseen constant for independent summands. Manuscript dated 2026-10-06; collection version 0.7.0; source snapshot f61bc638774bfa1165065462f98fe31efade30bf, 2026. URL https://github.com/epicycloids/independent-berry-esseen-bounds/blob/f61bc638774bfa1165065462f98fe31efade30bf/paper/independent-berry-esseen.pdf.

[3] Andrew C. Berry. The accuracy of the Gaussian approximation to the sum of independent variates. *Transactions of the American Mathematical Society*, 49(1):122–136, 1941. doi: 10.1090/S0002-9947-1941-0003498-3.

[4] Carl-Gustav Esseen. On the Liapunoff limit of error in the theory of probability. *Arkiv för Matematik, Astronomi och Fysik*, 28A(9):1–19, 1942.

[5] Carl-Gustav Esseen. Fourier analysis of distribution functions. A mathematical study of the Laplace–Gaussian law. *Acta Mathematica*, 77:1–125, 1945. doi: 10.1007/BF02392223.

[6] Carl-Gustav Esseen. A moment inequality with an application to the central limit theorem. *Scandinavian Actuarial Journal*, 1956(2):160–170, 1956. doi: 10.1080/03461238.1956.10414946.

[7] Hengzhi He and Guang Cheng. The Berry–Esseen constant conjecture is eventually true, 2026. URL https://arxiv.org/abs/2609.06358v3. Version 3, revised 30 September 2026.

[8] Yongxi Lin. Berry–Esseen bounds in Lean. GitHub repository, 2026. URL https://github.com/CoolRmal/BerryEsseen. Accompanying proof document: The i.i.d. Berry–Esseen constant is at most 0.423: a machine-checked proof, 13 September 2026, paper/upper-0423.pdf. Commit bfb22608ad18b4fa9fda8dd9a1ed2c2b6fbc29e7; accessed 27 October 2026.

[9] Håkan Prawitz. Limits for a distribution, if the characteristic function is given in a finite domain. *Scandinavian Actuarial Journal*, 1972(2):138–154, 1972. doi: 10.1080/03461238.1972.10404645.

[10] Kacper Rodziewicz and Bartosz Kołodziejek. Moment defects and sharp iid Berry–Esseen bounds. Hexagon preprint hexagon:2610.00147v3, 2026. URL https://hexagonmath.org/2610.00147v3. Working draft, version 3, 17 October 2026; accessed 27 October 2026.

[11] Jona Schulz. *The optimal Berry–Esseen constant in the binomial case*. PhD thesis, Universität Trier, 2016. URL https://ubt.opus.hbz-nrw.de/frontdoor/index/index/docId/732.

[12] Irina Shevtsova. On the absolute constants in the Berry–Esseen type inequalities for identically distributed summands, 2011. URL https://arxiv.org/abs/1111.6554v1.

[13] Irina Shevtsova. On the accuracy of the approximation of the complex exponent by the first terms of its Taylor expansion with applications. *Journal of Mathematical Analysis and Applications*, 418(1):185–210, 2014. doi: 10.1016/j.jmaa.2014.03.075. URL https://arxiv.org/abs/1301.2783v3.

[14] Irina G. Shevtsova. On the absolute constants in the Berry–Esseen inequality and its structural and nonuniform improvements. *Informatics and its Applications*, 7(1):124–125, 2013. URL https://www.mathnet.ru/eng/ia252. In Russian.

[15] Alexey Solovyev and Thomas C. Hales. Formal verification of nonlinear inequalities with Taylor interval approximations. In *NASA Formal Methods*, volume 7871 of *Lecture Notes in Computer Science*, pages 383–397, 2013. doi: 10.1007/978-3-642-38088-4_26.

[16] I. S. Tyurin. Refinement of the upper bounds of the constants in Lyapunov's theorem. *Russian Mathematical Surveys*, 65(3):586–588, 2010. doi: 10.1070/RM2010v065n03ABEH004688.

[17] Paul van Beek. An application of Fourier methods to the problem of sharpening the Berry–Esseen inequality. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 23: 187–196, 1972. doi: 10.1007/BF00536558.

[18] Haonan Xiao and Chunlin Li. A 0.4395 upper bound for the i.i.d. Berry–Esseen constant. GitHub repository, 2026. URL https://github.com/haonan-xiao/iid-berry-esseen. Accompanying paper: paper/berry-esseen-04395.pdf. Commit f00d781c8c42908bf4aef2739129442a0919fa38; accessed 27 October 2026.

[19] Vladimir M. Zolotarev. An absolute estimate of the remainder term in the central limit theorem. *Theory of Probability and its Applications*, 11(1):95–105, 1966. doi: 10.1137/1111005.

[20] Vladimir M. Zolotarev. A sharpening of the inequality of Berry–Esseen. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 8:332–342, 1967. doi: 10.1007/BF00531598.